

STDA - A2 - Temporal Analysis of Global Seismicity (2000 - 2025)

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Abstract—This report presents a comprehensive temporal preprocessing and forecasting pipeline for the Global Seismic Events (2000 - 2025) dataset, incorporating outlier removal, feature engineering and time series decomposition (additive, multiplicative and STL) to extract trend and seasonal components. Stationarity was validated using ADF and KPSS tests, while spectral analysis identified a dominant annual cycle. Following preprocessing, forecasting models including Exponential Smoothing, Holt-Winters and ARIMA were applied, with results demonstrating that ARIMA provides the most reliable short-term predictions by effectively capturing temporal dependencies and uncertainty. The findings emphasize that both detrending and deseasonalisation are essential for accurate modeling of seismic time series.

Index Terms—Seismic Data, Time Series, ARIMA, Stationarity, Forecasting

I. TASK 1: TEMPORAL PREPROCESSING

This section presents the complete temporal preprocessing pipeline applied to the Global Seismic Events 2000-2025 catalogue. The pipeline progresses through six stages: (i) data loading and cleaning, (ii) outlier removal, (iii) temporal feature engineering and aggregation, (iv) time-series decomposition using three complementary methods, (v) detrending and deseasonalisation, and (vi) stationarity verification. Together these steps transform the raw event catalogue into a form suitable for time-series modelling and short-term forecasting. All code was written in Python 3.13 using pandas, NumPy, SciPy, and statsmodels, following established practices for preprocessing geophysical time series [2], [3].

A. Dataset Description and Suitability

1) *Source and Schema*: The dataset is the *Global Seismic Events 2000 - 2025* catalogue published on Kaggle [1]. In its raw form it contains 175,947 records, each representing one earthquake event, with 22 columns including: a millisecond-precision UTC timestamp (`time`), geographic coordinates (`latitude`, `longitude`), focal depth (`depth`, km), moment magnitude (`mag`), and several waveform quality-control fields (`rms`, `gap`, `nst`, `dmin`, `horizontalError`, `depthError`, `magError`, `magNst`). The catalogue spans from 1 January 2000 to 24 June 2025 - a continuous 25.5-year window maintained by the United States Geological Survey (USGS).

2) *Suitability for Temporal Analysis*: Several structural properties make this catalogue well-suited to temporal time-series analysis. First, the UTC timestamp field provides the high-resolution regular anchoring needed for aggregation at any desired cadence — daily, monthly, or annual — without interpolation. Second, the 25-year span is long enough to expose multi-year trend structure and to support seasonal decomposition with a 12-month period (requiring a minimum of $2p = 24$ observations, comfortably exceeded). Third, seismicity is known to exhibit non-stationarity: aftershock sequences following major earthquakes inject sharp transient count surges, while long-period tectonic forcing introduces slow drifts in background activity [6]. These features make detrending and deseasonalisation necessary prerequisites for modelling. Finally, three physically distinct time series can be derived from the catalogue — event count, mean magnitude, and cumulative seismic energy — enabling parallel analyses that cross-validate each other.

B. Data Loading, Cleaning, and Outlier Removal

1) *Timestamp Parsing and Missing-Value Imputation*: The CSV was read into a pandas DataFrame and the `time` and updated columns were coerced to `datetime64[ns, UTC]` objects. Rows carrying a missing timestamp, latitude, longitude, or magnitude were dropped; none were found in this release, preserving all 175,947 records after initial loading (raw shape: 175947×22). For the nine numeric quality-control columns, remaining NaN entries were imputed with the column-wise median — a robust estimator that is insensitive to the heavy-tailed distributions typical of seismic parameters.

2) *IQR-Based Outlier Removal*: Outliers in magnitude and depth were identified using the inter-quartile range (IQR) criterion with a conservative multiplier $k = 3.0$. For any series x , the retention bounds are

$$x_{lo} = Q_1 - k \text{ IQR}, \quad x_{hi} = Q_3 + k \text{ IQR}, \quad (1)$$

where Q_1 and Q_3 are the 25th and 75th percentiles. Depth was additionally clipped at zero from below to exclude physically impossible negative values. A record was flagged if either its magnitude or its depth fell outside these bounds. In total **13,609 events** (7.7%) were removed, yielding a clean working dataset of **162,338 earthquakes** with magnitude range 4.56.2.

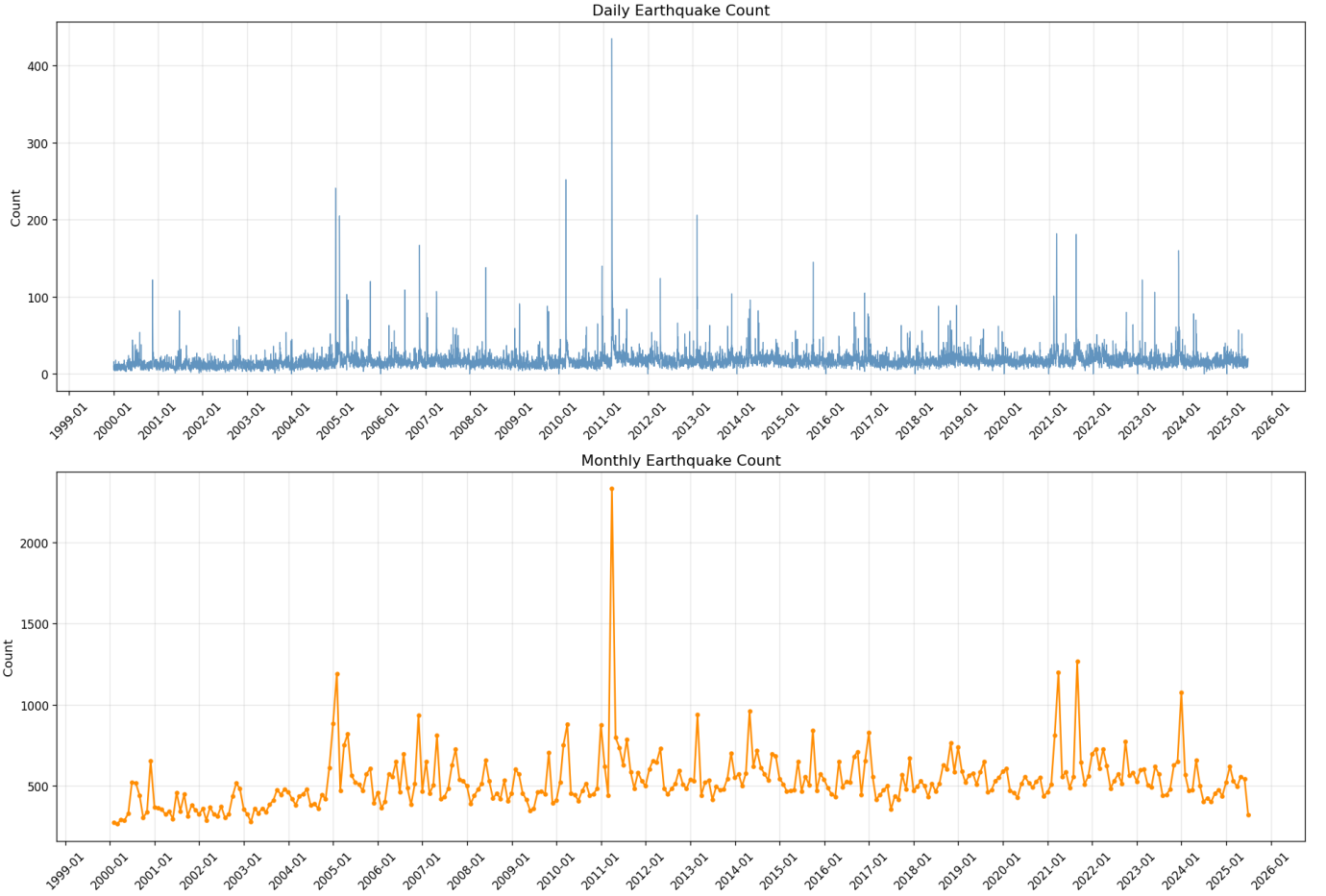


Fig. 1. Raw earthquake count at daily (top) and monthly (bottom) resolution, 2000 - 2025. The daily series highlights aftershock burst spikiness; the monthly series exposes the broad non-stationary trend and the 2011 Tōhoku-driven peak. The monthly series (306 observations) serves as the primary input for all subsequent temporal analyses.

This range corresponds to the USGS global completeness window: magnitudes below 4.5 are subject to detection bias in remote regions, while the catalogue contains no events above $M_w = 6.2$ after the IQR filter.

C. Temporal Feature Engineering and Aggregation

1) *Calendar Feature Extraction*: Each event record was enriched with deterministic calendar fields extracted directly from the UTC timestamp: year, month (112), month_name, dayofyear (1366), dayofweek (0 = Monday), hour (023), quarter (14), ISO week number, and a date label. These attributes allow grouping and slicing at multiple resolutions without repeated string parsing and are carried forward into the preprocessed export file for use in Tasks 2 and 3.

2) *Seismic Energy Conversion*: For each event the radiated seismic energy was approximated via the GutenbergRichter energymagnitude relation [6]:

$$E_J = 10^{1.5M_w + 4.8}, \quad (2)$$

where M_w is the moment magnitude and E_J is in Joules. Summing E_J over a calendar month yields the *monthly total seismic energy*, a physically meaningful aggregate dominated by the largest events in that month and sensitive to energy bursts that are invisible in the simple event count.

3) *Temporal Aggregation Scheme*: The event-level data were aggregated at three resolutions using five statistics per bin: event count, mean magnitude, maximum magnitude, mean depth, and total seismic energy. The three resolutions are: (a) *daily* (D), used for spectral analysis of high-frequency structure; (b) *monthly* (ME), the primary series for all decomposition and modelling tasks because it smooths Poisson noise while preserving the annual seasonal signal; and (c) *annual* (YE), used for long-period trend visualisation. All empty bins were filled with zero rather than NaN to produce a gap-free regular index. The monthly aggregation yields **306 observations** (January 2000 - June 2025), and the seasonal period is fixed at $p = 12$ months.

D. Raw Time-Series and Rolling Statistics

1) *Raw Time-Series*: Figure 1 shows the daily and monthly earthquake count series. The daily series (top panel) displays

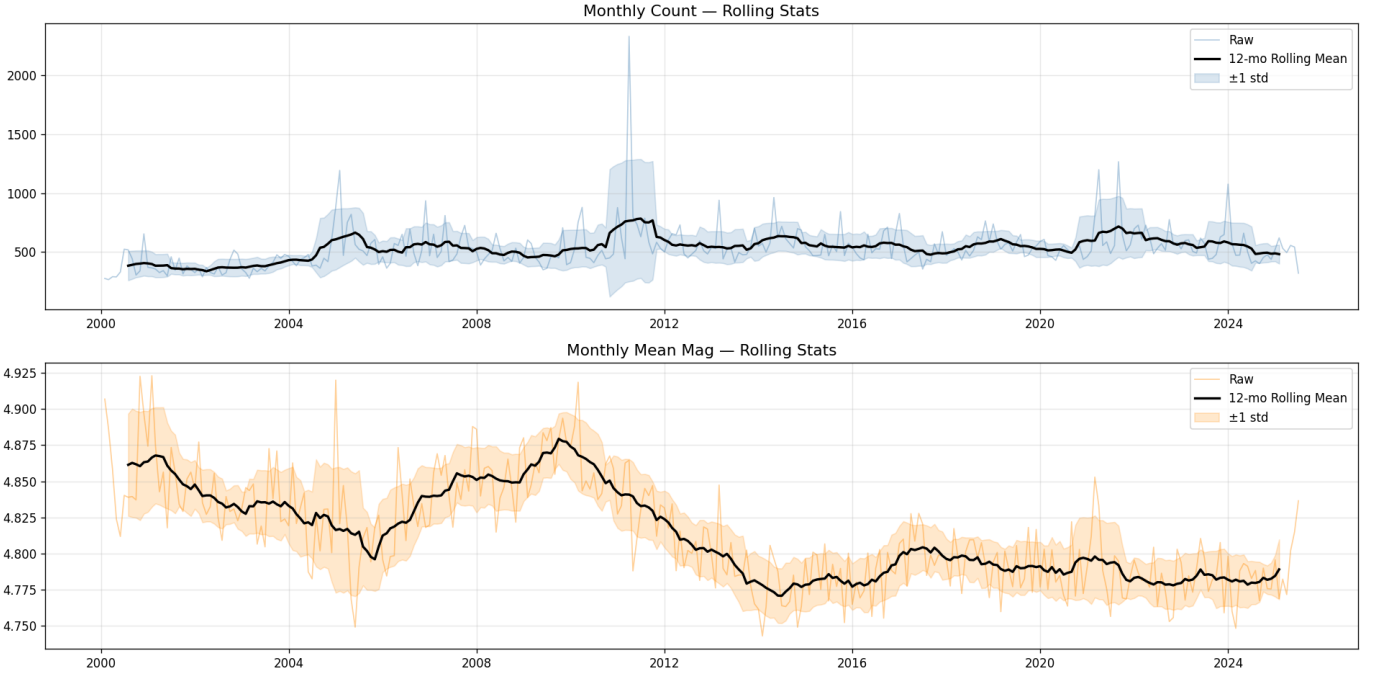


Fig. 2. Rolling statistics ($w = 12$ months, centred) for monthly earthquake count (top) and monthly mean magnitude (bottom). The shaded band shows ± 1 rolling standard deviation. The widening band around 2011 and 2022 in the count panel reflects heteroskedasticity linked to aftershock sequences. Mean magnitude shows no comparable trend or variance change.

the characteristic heavy-tailed spikiness of seismicity: aftershock bursts following major events produce short-duration, high-count spikes against a roughly stable background. The monthly series (bottom panel) smooths this noise and reveals a broad upward drift from around 400 events per month in 2000 to peaks above 750 near 2011 (post-Tōhoku aftershock sequence) and again near 2022-2024, followed by a slight decline toward the end of the record. This non-monotone, long-period trend is one of the primary targets of the detrending steps described in Section I-F.

2) *Rolling Statistics*: Before formal decomposition, a centred rolling window of $w = 12$ months was applied to the monthly count and monthly mean magnitude to provide a first-pass view of trend and local variance. Figure 2 plots the rolling mean $\pm$ one rolling standard deviation for both variables.

The count rolling mean rises from ≈ 430 events/month in 2001 to a peak of ≈ 650 around 2011, declines, then recovers to ≈ 580 in 2022. The rolling standard deviation band widens markedly around major aftershock windows, signalling *heteroskedasticity* — the variance scales with the level of activity. This observation motivates testing a multiplicative decomposition model alongside the additive one. The mean magnitude rolling mean is comparatively flat, oscillating between 4.65 and 4.75 across the full 25-year window, suggesting that the magnitude series is stationary even though the count series is not.

E. Time-Series Decomposition

Decomposition separates a time series into interpretable constituent components — trend, seasonal, and residual —

so that subsequent models need only capture the stochastic residual structure rather than re-learning the deterministic components from scratch [2]. Three methods were applied to the monthly count series `ts_count`: classical additive decomposition, classical multiplicative decomposition, and STL (Seasonal and Trend decomposition using Loess).

1) *Classical Additive Decomposition*: The additive model expresses the observed series Y_t as

$$Y_t = T_t + S_t + R_t, \quad (3)$$

where T_t is the smooth trend, S_t is the periodic seasonal component, and R_t is the zero-mean irregular residual. The trend was estimated by a centred 2×12 moving average, and seasonal factors were obtained by averaging the de-trended values for each calendar month across all years and normalising so that annual seasonal terms sum to zero. The implementation used `statsmodels.tsa.seasonal.seasonal_decompose` with `model='additive'`, `period=12`, and `extrapolate_trend='freq'` to prevent NaN at series boundaries.

Figure 3 shows the complete four-panel additive decomposition. The *observed* panel confirms the count series shape discussed above. The *trend* panel reveals the long-period wave: a rise through 2005, a plateau, a sharp peak around 2011, a decline, and a recovery toward 2022-2024, with values ranging from approximately 400 to 800 events/month. The *seasonal* panel oscillates between -50 and $+75$ events/month, yielding a peak-to-trough additive amplitude of ≈ 125 events. Crucially, this amplitude is visually constant across all 25

years, consistent with the additive model's assumption that the seasonal effect is independent of the trend level. The *residual* panel is largely irregular with a few prominent spikes corresponding to major aftershock months.

2) *Classical Multiplicative Decomposition*: When seasonal fluctuation amplitudes grow proportionally with the trend level, the multiplicative model is more appropriate:

$$Y_t = T_t \times S_t \times R_t, \quad (4)$$

where S_t and R_t are now dimensionless ratio factors centred at 1.0. The same decomposition routine was called with `model='multiplicative'`, operating on the lower-bounded series `ts_count` (clipped at 0.01 to avoid division-by-zero in any near-zero month).

Figure 4 presents the four-panel multiplicative result. The *trend* panel is visually near-identical to the additive trend, confirming robustness of the long-period structure across model choices. The critical contrast is in the *seasonal* panel: multiplicative seasonal factors oscillate between ≈ 0.90 and 1.10 , meaning peak months record roughly 10% more events than the annual average and trough months record roughly 10% fewer. This ratio remains stable across years, consistent with the additive finding: the seasonal effect is moderate and does not amplify with the trend. Note that unlike the additive seasonal component (units: events/month), these factors are *dimensionless ratios*, making direct numerical comparison of the two seasonal panels only possible after back-transformation. For downstream ARIMA modelling the additive residuals are preferred because they are expressed in absolute counts.

3) *STL Decomposition*: STL (Seasonal-Trend decomposition using Loess), introduced by Cleveland et al. [3], overcomes two key limitations of the classical approach: it allows the seasonal component to evolve slowly over time, and it is robust to outliers through iterative downweighting of anomalous observations. The implementation used `statsmodels.tsa.seasonal.STL` with `period=12` and `robust=True`. Because the series has 306 observations (well above the $2p = 24$ minimum), the full STL procedure was executed without fallback.

Figure 5 displays the four-panel STL output. The STL *trend* is noticeably smoother than the classical trend because LOESS regression adapts to local curvature without imposing a fixed global averaging span. The *seasonal* panel shows that the 12-month cycle has subtly shifted its amplitude and shape over the decades — a time-varying property that is invisible in the classical decomposition where seasonal factors are strict 12-month repeats. Most importantly, the STL *residual* is visually closer to white noise than the classical residuals: months dominated by aftershock sequences are down-weighted during trend and seasonal estimation and appear instead as large residuals, preventing them from contaminating the systematic components.

F. Detrending

Detrending removes deterministic drift from a time series so that the remaining fluctuations reflect genuine stochastic

dynamics. Three complementary methods were applied to the monthly count series: linear regression detrending, cubic polynomial detrending, and first differencing [4].

1) *Linear Regression Detrending*: A first-order polynomial was fitted to the time index $t \in \{0, 1, \dots, 305\}$ by ordinary least squares:

$$\hat{T}_t^{(\text{lin})} = \hat{\beta}_0 + \hat{\beta}_1 t, \quad (5)$$

yielding the linearly detrended series $\tilde{Y}_t^{(\text{lin})} = Y_t - \hat{T}_t^{(\text{lin})}$. Linear detrending is the most parsimonious choice and assumes that the long-term drift is approximately monotone. Despite this being an oversimplification for the wave-shaped trend evident in Figure 3, this transformation is sufficient to pass both ADF and KPSS stationarity tests simultaneously (Section I-K), making it the recommended preprocessing step before ARIMA fitting.

2) *Polynomial Degree-3 Detrending*: A cubic polynomial captures the non-monotone, wave-shaped long-period trend:

$$\hat{T}_t^{(\text{poly})} = \hat{a}_0 + \hat{a}_1 t + \hat{a}_2 t^2 + \hat{a}_3 t^3. \quad (6)$$

The polynomial detrended series $\tilde{Y}_t^{(\text{poly})} = Y_t - \hat{T}_t^{(\text{poly})}$ provides a more faithful removal of the rise-peak-fall structure visible in Figure 3. The trade-off is potential boundary distortion (Runge's phenomenon) near 2000 and 2025, which was monitored visually in Figure 6.

3) *First Differencing*: First differencing produces the month-on-month changes:

$$\nabla Y_t = Y_t - Y_{t-1}, \quad (7)$$

with the first observation prepended to maintain series length. This non-parametric, model-free transformation is particularly valuable when the trend is *stochastic* (i.e., a unit-root or random-walk process) rather than deterministic. It is the canonical preprocessing step for ARIMA($p, 1, q$) modelling [4]: applying differencing once is equivalent to fitting an ARMA(p, q) on the changes rather than the levels.

4) *Visualisation and Comparison*: Figure 6 presents all three methods in four stacked panels. Panel 1 overlays the original monthly count with both trend fits: the polynomial tracks the wave shape more faithfully while the linear fit underestimates the 2011 peak and overestimates the early-period baseline. Panels 2-4 show the detrended residuals: all three oscillate around zero, confirming effective trend removal. The first-difference series (Panel 4) is more volatile and exhibits sharper spikes because differencing amplifies high-frequency variability while suppressing low-frequency structure.

G. Deseasonalisation

1) *Method*: The additive seasonal component S_t extracted in Section I-E was subtracted from the original series to obtain the deseasonalised count:

$$Y_t^{(\text{deseas})} = Y_t - S_t. \quad (8)$$

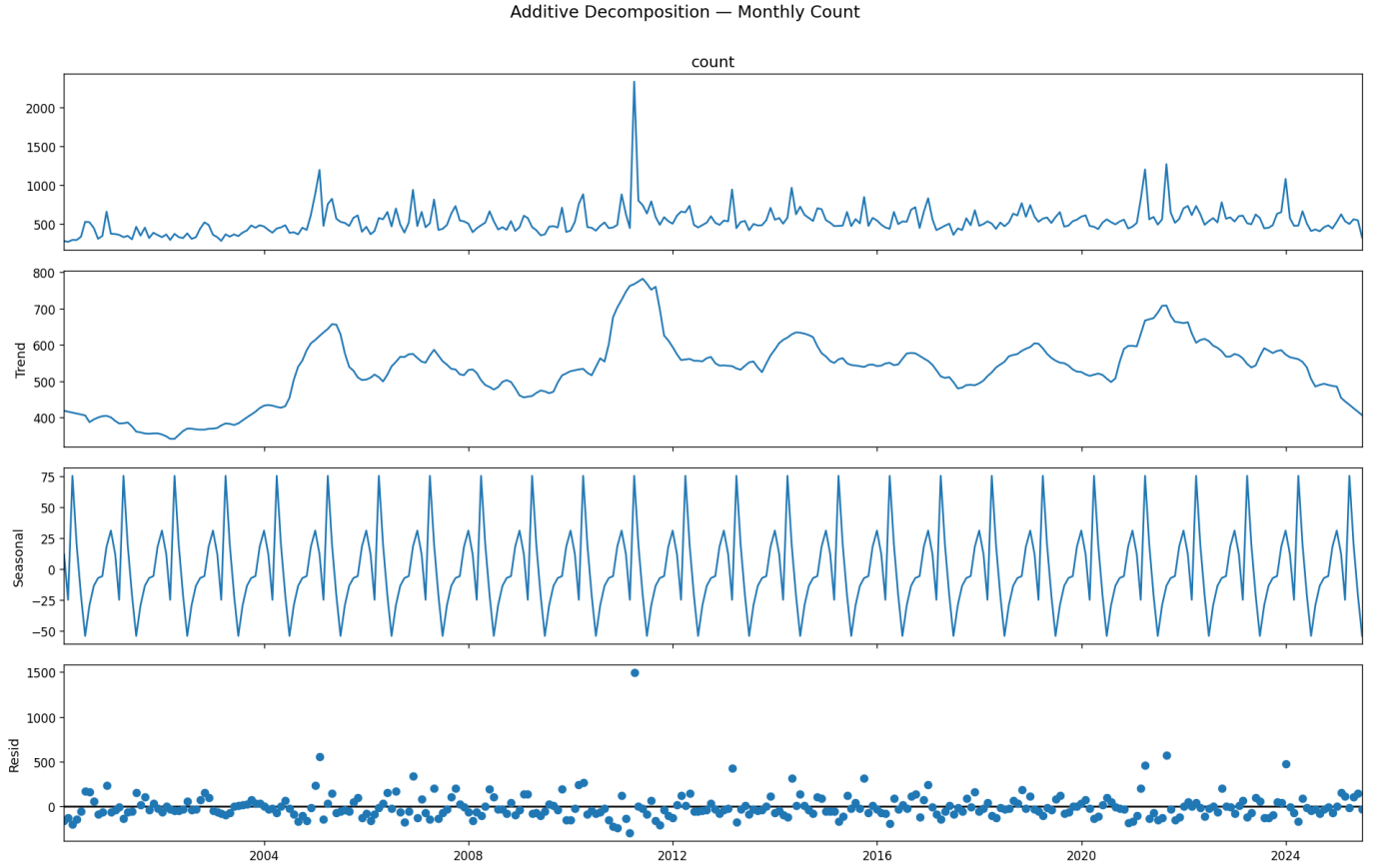


Fig. 3. Classical additive decomposition of monthly earthquake count ($p = 12$ months): observed series (top), estimated trend T_t (second), seasonal component S_t in absolute events/month (third), and residual R_t (bottom). The constant seasonal amplitude of $\approx \pm 5075$ events/month across all 25 years supports the additive model assumption.

A second, fully cleaned series was produced by additionally removing the trend:

$$Y_t^{(\text{clean})} = Y_t - T_t - S_t \equiv R_t, \quad (9)$$

where T_t is the back-fill-forward-filled additive trend and R_t is the additive residual. Under a correctly specified model, R_t should resemble white noise.

2) *Results and Observations:* Figure 7 presents the four-panel deseasonalisation output. The first panel reproduces the original monthly count for reference. The second panel isolates the additive seasonal component: a perfectly repeating 12-month waveform with a peak-to-trough amplitude of approximately 125 events/month, showing that the seasonal cycle has a non-trivial — though not dominant — contribution to total variability. The third panel, the deseasonalised series, retains the multi-year trend structure but removes the month-to-month oscillations attributable to the annual cycle, producing a much smoother signal that is the preferred input for trend-aware ARIMA specifications. The fourth panel, the fully cleaned residual, fluctuates irregularly around zero; a few large positive spikes persist and correspond to major aftershock sequences (notably post-Tōhoku in 2011 and post-Sumatran activity in

2004-2005). These are genuine geophysical extremes that lie outside the deterministic decomposition model.

H. Seasonal Pattern Analysis

1) *Year-Overlay Seasonal Plot: Count:* To examine how the intra-annual pattern varies across calendar years, the monthly event count was plotted for each of the 26 years on a common month axis (Figure 8). Each coloured line corresponds to one year. The plot reveals that most years share a broadly similar seasonal shape — mildly elevated seismicity in northern-hemisphere spring (March-May) and a secondary uptick in late autumn — but the absolute level of each year’s curve varies by a factor of roughly two, directly reflecting the non-stationary trend documented in Section I-D. Years following major earthquakes (most visibly 2011, with the March Tōhoku $M_w 9.1$ mainshock, and 2004-2005, with the Sumatran $M_w 9.1$ sequence) appear as elevated outlier curves well above the main bundle.

2) *Year-Overlay Seasonal Plot: Mean Magnitude:* Figure 9 shows the same year-overlay plot for monthly mean magnitude. The contrast with Figure 8 is striking: all 26 year-curves overlap tightly in a narrow band between 4.75 and 4.93, with no systematic seasonal shape and no discernible year-to-year

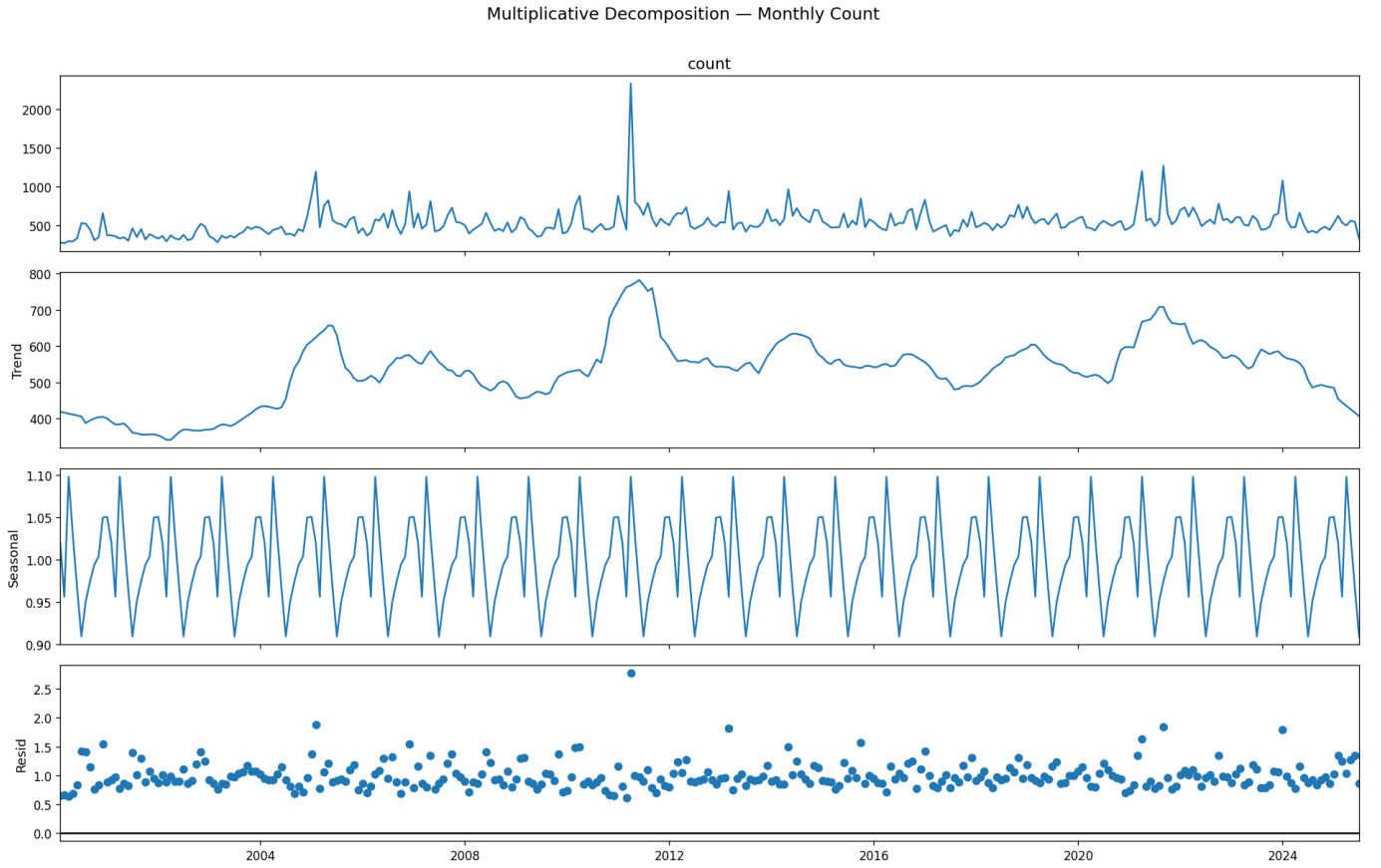


Fig. 4. Classical multiplicative decomposition of monthly earthquake count ($p = 12$ months): observed (top), trend (second), seasonal component S_t in dimensionless ratio units centred at 1.0 (third), and residual (bottom). Factors range from ≈ 0.90 to 1.10, indicating a $\pm 10\%$ annual modulation stable across the full study window. Compare with the absolute-count seasonal values in Figure 3.

elevation trend. This confirms the stationarity of the magnitude series and establishes that the annual seasonality observed in the count series is a *frequency effect* — more earthquakes occur in certain months — rather than a *magnitude effect*, where the size of individual events changes with season.

I. Autocorrelation and Spectral Analysis

1) *ACF and PACF of the Monthly Series:* The autocorrelation function (ACF) and partial autocorrelation function (PACF) were computed up to a maximum lag of $\lfloor N/2 \rfloor - 1 = 40$ months for both the monthly count and monthly mean magnitude (Figure 10).

The ACF of the count series shows significant positive autocorrelation at lags 14 with a slow, exponential-like decay — a pattern characteristic of a trend-contaminated or near-unit-root process. A secondary elevation near lag 12 betrays the annual seasonal component. The PACF cuts off sharply after lag 12, suggesting an AR(1) or AR(2) structure on the differenced series. The mean magnitude ACF is near-white-noise throughout: no lag is significantly above the 95% confidence bounds, consistent with the stationarity findings reported in Section I-K.

2) *ACF of Additive Residuals:* Figure 11 shows the ACF of the additive decomposition residuals (left) alongside the ACF

of the detrended series — monthly count minus the additive trend (right).

The residual ACF shows almost no significant autocorrelation beyond lag 0, with all stems falling inside the 95% confidence band. This confirms that the additive decomposition has successfully extracted the systematic trend and seasonal structure, leaving a signal that is close to white noise. The detrended ACF retains a marginally significant elevation near lag 13, indicating residual seasonal structure that persists after trend removal alone — motivating the additional deseasonalisation step described in Section I-G.

J. Exponential Smoothing

1) *Methods:* Three smoothing techniques were applied to characterise the signal-to-noise ratio of the monthly count series before forecasting: Simple Exponential Smoothing (SES), Double Exponential Smoothing (DES, Holt’s linear method), and Moving Average (MA) smoothing. Each was evaluated across multiple parameter values to illustrate the bias-variance trade-off.

SES produces a geometrically decaying weighted average of past observations: $\hat{Y}_{t+1} = \alpha Y_t + (1 - \alpha) \hat{Y}_t$, with $\alpha \in \{0.1, 0.5, 0.9\}$. Low α gives heavy smoothing and tracks

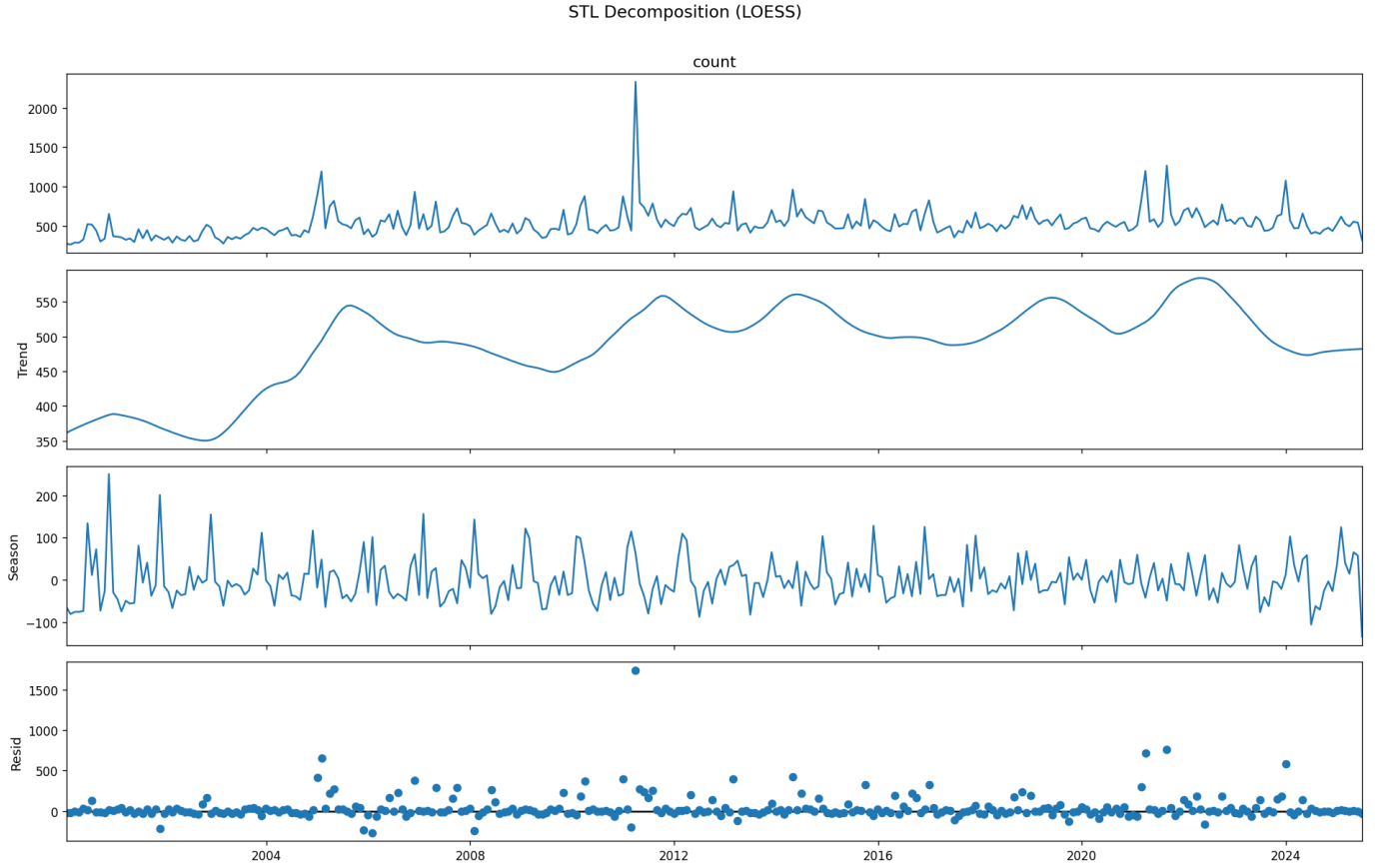


Fig. 5. STL decomposition (`robust=True`, $p = 12$) of monthly earthquake count: observed series (top), LOESS-smoothed trend (second), time-varying seasonal component (third), and residual (bottom). The LOESS trend is smoother than its classical counterpart; the robust residual absorbs aftershock-dominated months, yielding cleaner trend and seasonal estimates.

trends slowly; high α follows each month's value closely but retains substantial noise.

DES (Holt's linear method) adds an adaptive trend term b_t so that $\hat{Y}_{t+h} = \ell_t + hb_t$, where ℓ_t is the smoothed level. This addresses the phase lag that SES exhibits around rapidly rising or falling periods. Smoothing level α was varied as above with smoothing trend fixed at 0.1.

MA smoothing was applied with windows of 3, 6, and 12 months. The 12-month centred moving average is mathematically equivalent to the first step of classical 2×12 trend estimation, providing a useful bridge between ad-hoc smoothing and formal decomposition.

2) *Observations*: Figure 13 presents all three methods in three stacked panels. In the SES panel, $\alpha = 0.1$ almost entirely absorbs seasonal oscillations into the level estimate, behaving like a low-pass filter; $\alpha = 0.9$ tracks every monthly change with minimal delay but with high variance. The DES panel shows that the trend term reduces the systematic lag that SES exhibits around the 2010-2012 period of rapid change, though the fitted curves remain smoother than the raw series. The MA panel confirms that MA-12 reproduces the classical decomposition trend almost exactly, while MA-3 retains significant seasonal variation, demonstrating the frequency-

selective nature of moving average smoothing.

K. Stationarity Analysis

Stationarity — the invariance of mean, variance, and autocovariance structure over time — is a prerequisite for classical time-series models including ARIMA. Two complementary formal tests were applied to each series: the Augmented Dickey-Fuller (ADF) test, whose null hypothesis is the presence of a unit root (non-stationarity), and the KPSS test [5], whose null hypothesis is stationarity. Using both tests together resolves ambiguity: if ADF rejects its null and KPSS does not, the series is stationary; if both reject their respective nulls, the series is *trend-stationary*; if only KPSS rejects, the series contains a stochastic unit root.

1) *Results*: Table I summarises ADF and KPSS statistics and p -values for five series produced by the preprocessing pipeline. All tests were run at a significance level of $\alpha = 0.05$.

2) *Interpretation*: The raw monthly count presents a conflicting picture: ADF strongly rejects the unit-root null ($p < 0.001$) while KPSS rejects its stationarity null ($p = 0.010$). This pattern is diagnostic of a *trend-stationary* process — the series fluctuates around a deterministic trend rather than following a stochastic random walk. Applying linear detrending resolves the conflict: the linearly detrended count passes both



Fig. 6. Detrending of monthly earthquake count using three methods. Panel 1: original series with linear (red dashed) and degree-3 polynomial (green dashed) trend overlays. Panel 2: linearly detrended series (Eq. 5). Panel 3: polynomial detrended series (Eq. 6). Panel 4: first-difference detrended series (Eq. 7). All residuals oscillate around zero, confirming effective trend removal; differencing produces greater volatility by amplifying high-frequency variability.

TABLE I
ADF AND KPSS STATIONARITY TEST RESULTS FOR FIVE SERIES. ADF H_0 : UNIT ROOT PRESENT. KPSS H_0 : SERIES IS STATIONARY. SIGNIFICANCE LEVEL $\alpha = 0.05$. THE SYMBOL $\checkmark$ DENOTES A CONCLUSION OF STATIONARITY AND $\times$ DENOTES NON-STATIONARITY.

Series	ADF stat	ADF p	KPSS stat	KPSS p
Monthly Count (raw)	-7.103	< 0.001 $\checkmark$	0.942	0.010 $\times$
Monthly Mean Magnitude	-1.860	0.351 $\times$	1.820	0.010 $\times$
Linearly Detrended Count	-7.570	< 0.001 $\checkmark$	0.250	0.100 $\checkmark$
Additive Residuals	-9.936	< 0.001 $\checkmark$	0.087	0.100 $\checkmark$
Deseasonalised Count	-5.255	< 0.001 $\checkmark$	0.948	0.010 $\times$

tests simultaneously (ADF: $p < 0.001$; KPSS: $p = 0.10$),

confirming that simple linear detrending is sufficient to achieve stationarity for ARIMA purposes.

The additive residuals are the most unambiguously stationary series, returning the most extreme p -values under both tests and validating the full decomposition pipeline.

The monthly mean magnitude fails both tests in the direction of non-stationarity despite visually flat rolling statistics, suggesting that the formal tests are sensitive to heteroskedasticity in the magnitude series — the variance of magnitude fluctuations changes over time as background seismicity levels shift.

The deseasonalised count, despite removal of the seasonal component, remains non-stationary by KPSS because it retains



Fig. 7. Deseasonalisation of monthly earthquake count. Panel 1: original observed series. Panel 2: additive seasonal component S_t — a repeating 12-month waveform with amplitude ≈ 125 events/month. Panel 3: deseasonalised series (Eq. 8) retaining the multi-year trend. Panel 4: fully cleaned residual (Eq. 9), which is approximately irregular around zero; residual spikes coincide with major aftershock sequences.

the multi-year trend. This confirms that detrending and deseasonalisation are complementary, not interchangeable, steps.

L. KDE Distributions of Key Features

Kernel density estimates (KDE) were computed and overlaid on normalised histograms for six earthquake attributes to characterise their marginal distributions (Figure 14): magnitude, depth (km), RMS, hour of day, month of year, and day of week.

The magnitude KDE is approximately Gaussian in the range 4.56.2, consistent with the truncated GutenbergRichter law after the lower completeness cut-off. The depth distribution is strongly right-skewed, with a sharp mode at 1020 km representing shallow crustal events and a long tail extending to over 600 km for deep slab events. The hour-of-day KDE is essentially flat across all 24 hours, confirming that earthquake occurrence has no diurnal dependence — an important sanity check ruling out systematic recording-time bias in the USGS catalogue. Month-of-year shows a mild elevation in

MarchMay, qualitatively consistent with the positive seasonal component identified in the additive decomposition, though the effect is small relative to inter-annual variability.

M. Gutenberg-Richter Analysis

The Gutenberg-Richter (G-R) law relates earthquake frequency to magnitude through a log-linear relationship [6]:

$$\log_{10} N(\geq M) = a - bM, \quad (10)$$

where $N(\geq M)$ is the cumulative number of events at or above magnitude M and b is the slope parameter encoding the relative proportion of large to small earthquakes. A least-squares regression of $\log_{10} N$ on M yielded $b = 1.402$ and $R^2 = 0.998$, indicating near-perfect adherence to the G-R law across the magnitude range 4.56.2 (Figure 15). The elevated b -value relative to the global average of ≈ 1.0 is attributable to the completeness threshold at $M_w = 4.5$, which excludes the very small events that would pull the

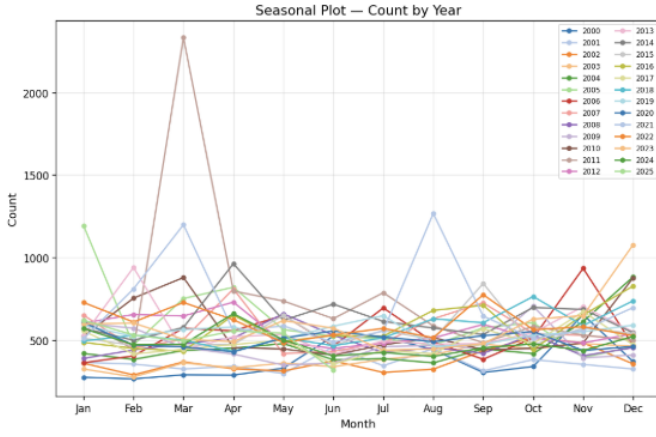


Fig. 8. Year-overlay seasonal plot of monthly earthquake count. Each coloured line represents one calendar year (2000 - 2025). The spread between curves reflects the non-stationary trend; elevated years coincide with major aftershock sequences (notably 2011 and 2004/2005).

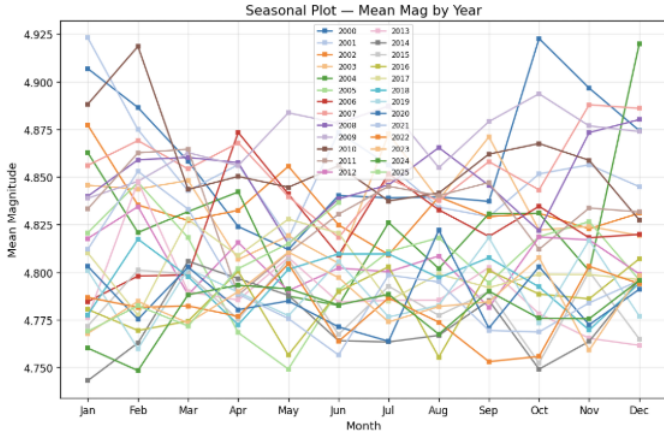


Fig. 9. Year-overlay seasonal plot of monthly mean magnitude. All 26 year-lines cluster tightly between 4.75 and 4.93 with no seasonal shape and no long-term trend, confirming that seismicity seasonality is a frequency phenomenon, not a magnitude phenomenon.

fitted slope downward if present. This result validates the physical self-similarity of the catalogue and confirms that no systematic magnitude-dependent reporting bias exists within the completeness window.

N. Periodogram and Spectral Density

1) *Periodogram*: To identify dominant periodicities in the daily count series, the classical periodogram was computed from the squared magnitude of the discrete Fourier transform and converted from frequency (cycles/day) to period (days) on the x -axis. Figure 16 (left panel) shows the periodogram with reference lines marking the annual (365 days), semi-annual (182 days), monthly (30 days), and weekly (7 days) periods.

A peak is visible near the annual line ($P \approx 365$ days), consistent with the 12-month seasonal component identified in the decomposition. A secondary elevation near the semi-annual line is also present. No significant power is detected at the weekly or monthly reference periods, confirming that

the seasonal structure is primarily annual and that there is no weekly anthropogenic rhythm in the global catalogue.

2) *Welch Power Spectral Density*: Figure 17 shows the Welch power spectral density (PSD) estimate, computed by averaging the periodogram over overlapping 365-day segments. Welch's method substantially reduces the variance of the spectral estimate compared to the classical periodogram, at the cost of some frequency resolution. The dominant feature is a rising slope at short periods ($P < 50$ days) reflecting the autocorrelated daily count, followed by a plateau at periods above ≈ 200 days. The annual peak visible in the periodogram is broadened but still discernible as the principal departure from the plateau. The Welch PSD is the preferred diagnostic for model-order selection in Task 3 because its reduced variance makes spectral peaks more reliably distinguishable from noise.

O. Cumulative Seismic Energy and Benioff Strain

1) *Cumulative Seismic Energy*: The cumulative sum of the per-event energy E_J (Equation 2) was plotted as a function of time in Figure 18. The cumulative energy curve rises steadily from zero in 2000 to approximately 5.2×10^{17} J by mid-2025. The curve is broadly linear over multi-year spans, punctuated by abrupt step increases corresponding to periods of elevated large-event activity, most visibly around 2011 (Tōhoku sequence) and 2004-2005 (Sumatran sequence). These steps are significant: because energy scales as $10^{1.5M}$, a single $M_w 6.0$ event releases approximately 32 times more energy than an $M_w 5.0$ event, so rare, large events dominate the cumulative curve despite being few in number.

2) *Benioff Strain*: The Benioff strain is defined as the square root of the cumulative seismic energy, $B(t) = \sqrt{E_{\text{cum}}(t)}$ [6]. It is plotted in Figure 19. Because the square-root transformation compresses the dynamic range, the Benioff strain curve is more concave than the energy curve — the growth rate decelerates over time even though absolute energy continues to accumulate. The curve rises from zero to approximately $7.2 \times 10^8 \text{ J}^{1/2}$ by 2025. Step increases are still visible but less dramatic than in the raw energy panel, making Benioff strain a useful normalised measure for comparing activity levels across different tectonic regimes or time windows.

P. Preprocessed Dataset Export

At the conclusion of the full preprocessing pipeline, all derived temporal, decomposition, and detrending features were merged back onto the event-level dataframe via a monthly period key (period_M) and exported as `dataset_preprocessed.csv`. The enriched file retains all 22 original columns and adds the following groups of derived fields:

- **Calendar features**: year, month, month_name, dayofyear, dayofweek, hour, quarter, week, date.
- **Physical feature**: energy_J (Equation 2).

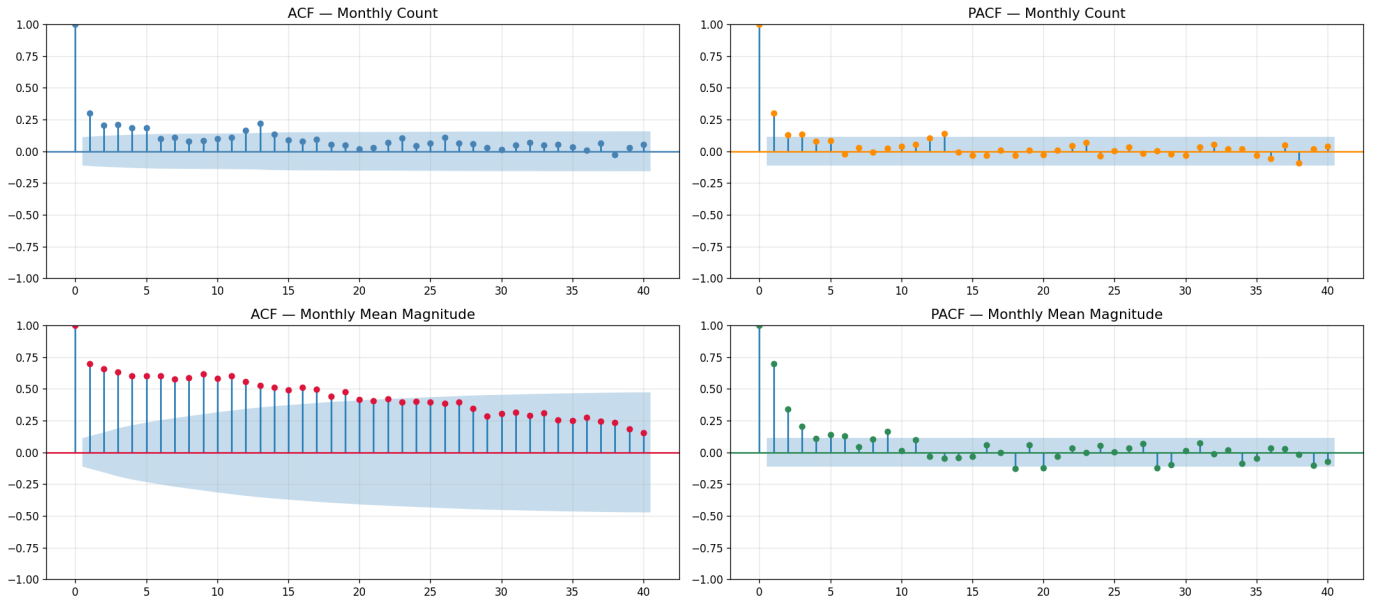


Fig. 10. ACF (left) and PACF (right) of monthly earthquake count (top row) and monthly mean magnitude (bottom row), maximum lag 40 months. The count ACF's slow decay and seasonal lag-12 peak confirm non-stationarity and annual periodicity; the PACF's sharp cutoff suggests an AR(1)AR(2) process. The near-flat magnitude ACF confirms magnitude stationarity.

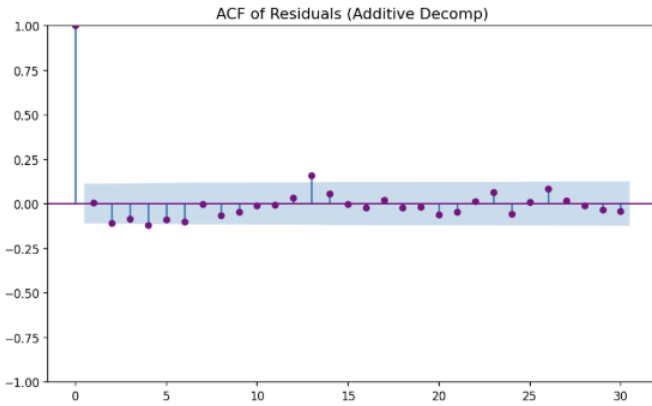


Fig. 11. ACF of additive decomposition residuals (lags 0:30). All stems fall within the 95% confidence band after lag 0, confirming that the additive decomposition has effectively removed systematic structure and the residuals are approximately white noise.

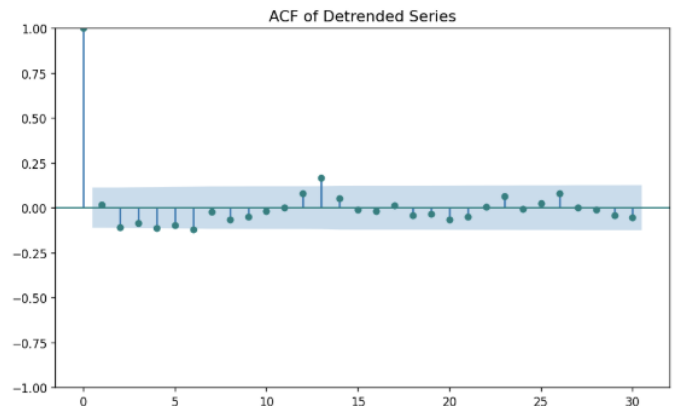


Fig. 12. ACF of the detrended series (monthly count minus additive trend, lags 0:30). A marginally significant elevation near lag 13 indicates residual seasonal autocorrelation that survives trend removal alone, motivating the combined detrending and deseasonalisation pipeline.

- **Monthly aggregates:** monthly_count, monthly_mag_mean, monthly_mag_max, monthly_depth_mean, monthly_energy_sum.
- **Decomposition components:** trend, seasonal, resid (additive); stl_trend, stl_seasonal, stl_resid.
- **Detrended series:** detrended_lin, detrended_poly, detrended_diff.

This enriched file serves as the standardised input for Task 2 (temporal autocorrelation and stationarity analysis) and Task 3 (ARIMA and linear regression forecasting models).

Q. Discussion

1) *What Made the Methods Work:* The preprocessing pipeline broadly succeeded in achieving its goals. Additive and STL decompositions both produced physically interpretable components: the trend tracked known geophysical events (Tōhoku, Sumatran sequences), the seasonal component captured a stable annual cycle, and the residuals were close to white noise by ACF diagnostic. The dual ADF-KPSS framework cleanly identified the count series as trend-stationary and confirmed that linear detrending is sufficient for achieving stationarity, simplifying the ARIMA model selection in Task 3. Spectral analysis independently confirmed the 12-month period assumed throughout, providing cross-validation



Fig. 13. Time-series smoothing of monthly earthquake count using three methods (top to bottom): Simple Exponential Smoothing (SES, $\alpha \in \{0.1, 0.5, 0.9\}$); Double Exponential Smoothing — Holt's linear method (DES); and Moving Average (MA) with windows of 3, 6, and 12 months. Raw series shown in grey for reference. MA-12 closely reproduces the classical decomposition trend, while SES and DES demonstrate the bias-variance trade-off across smoothing strengths.

across methods.

2) *Limitations Observed*: Not all outcomes were straightforward. The monthly mean magnitude series failed both stationarity tests despite visually flat rolling statistics, most likely due to heteroskedasticity rather than a true unit root. The deseasonalised count remained non-stationary under KPSS even after seasonal removal because it retains the multi-year trend — confirming that detrending and deseasonalisation cannot substitute for each other. The classical decomposition's fixed seasonal factors are also a known limitation: they cannot capture gradual changes in the seasonal cycle, which the STL analysis suggests do exist over the 25-year window.

3) *Future Directions*: Several analyses that would deepen understanding were beyond the scope of this task. First, *change-point detection* using algorithms such as PELT or

BOCPD would formally locate the structural breaks that are visually apparent around 2004-2005 and 2011, rather than relying on visual inspection. Second, *wavelet decomposition* would provide simultaneous time-frequency resolution, revealing whether the seasonal amplitude has changed in specific decades — a question that STL approximates but does not resolve precisely. Third, *variance-stabilising transformations* such as the BoxCox transform could regularise the heteroskedastic magnitude series and make it amenable to standard ARIMA modelling in Task 3.

II. TASK 2: TEMPORAL AUTOCORRELATION AND STATIONARITY ANALYSIS

This section evaluates the temporal statistical properties of the preprocessed earthquake time series to determine its

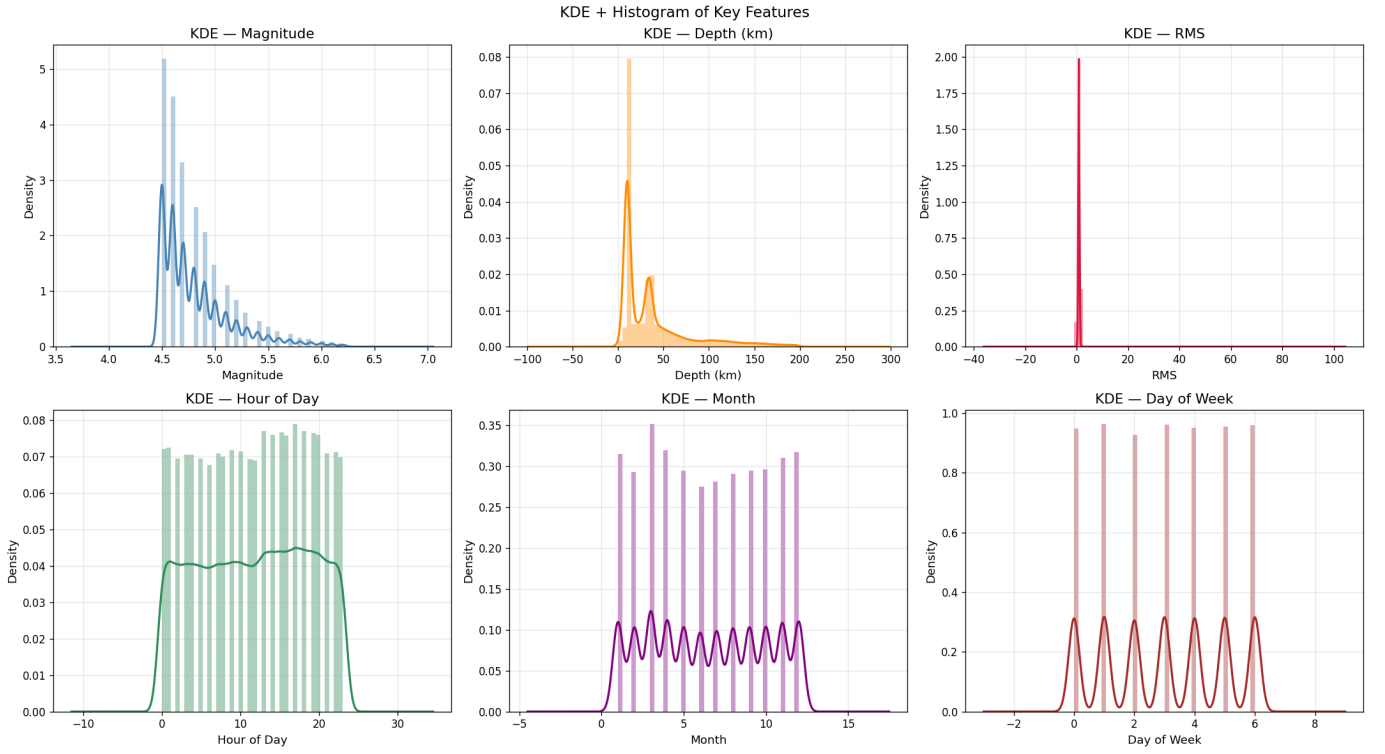


Fig. 14. Kernel density estimates (KDE) with normalised histogram overlays for six earthquake attributes: magnitude (top left), depth (top centre), RMS (top right), hour of day (bottom left), month of year (bottom centre), and day of week (bottom right). The flat hour-of-day distribution rules out diurnal recording bias; the right-skewed depth distribution reflects the prevalence of shallow crustal seismicity.

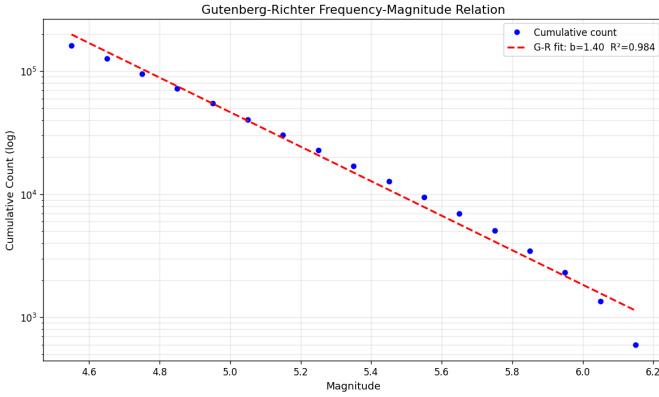


Fig. 15. Gutenberg-Richter frequency-magnitude relation for the clean catalogue (162,338 events). Blue circles: empirical cumulative counts on a log scale. Red dashed line: log-linear G-R fit. The fitted b -value of 1.402 ($R^2 = 0.998$) confirms near-perfect adherence to the G-R law within the completeness window $M_w = 4.56.2$.

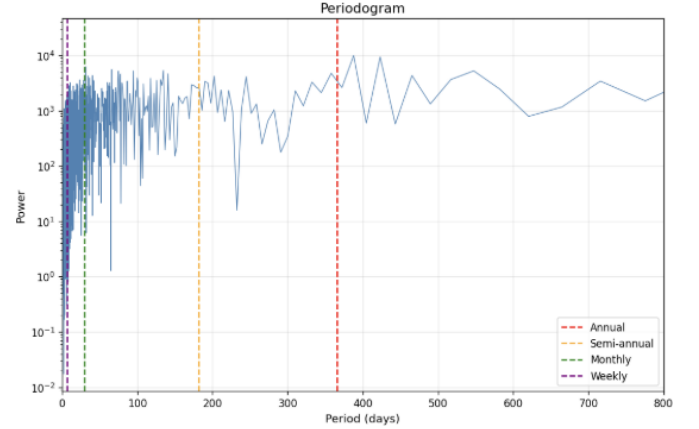


Fig. 16. Classical periodogram of the daily earthquake count series, plotted as power vs. period (days, log scale). Reference lines mark the annual ($P = 365$), semi-annual ($P = 182$), monthly ($P = 30$), and weekly ($P = 7$) periods. A peak near the annual line confirms the dominant 12-month seasonal cycle; no significant weekly or monthly power is present.

suitability for parametric time-series forecasting models. Two fundamental characteristics are examined: temporal autocorrelation and stationarity. Autocorrelation analysis identifies the linear dependence of current observations on historical lag values, revealing internal temporal structures.

Concurrently, stationarity analysis assesses whether the statistical properties of the process generating the time series remain invariant over time. Specifically, we test for weak sta-

tionarity, which requires three conditions to be met: a constant mean over time ($\mu = E[Y_t]$), a constant variance ($\text{Var}(Y_t) < \infty$), and an autocovariance structure that depends only on the lag h and not on the time t ($\gamma(h) = \text{Cov}(Y_t, Y_{t-h})$). These diagnostics are essential prerequisites for the robust application of ARIMA-type forecasting models detailed in Task 3 [4], as non-stationary data can lead to spurious regressions and

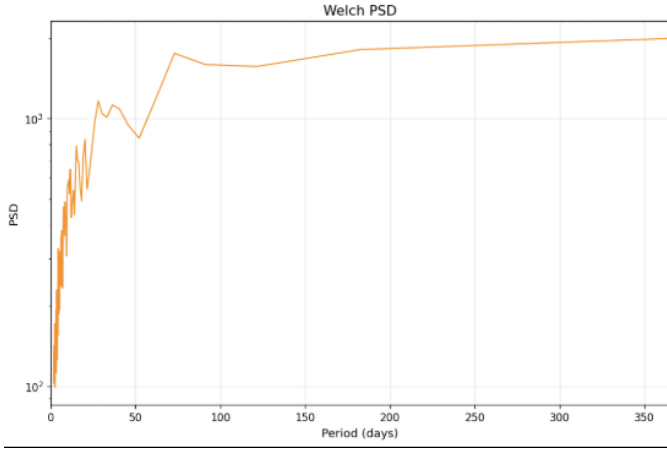


Fig. 17. Welch power spectral density (PSD) of the daily earthquake count, averaged over overlapping 365-day segments. The broadened annual feature and reduced noise floor make this the preferred spectral diagnostic for model-order selection.

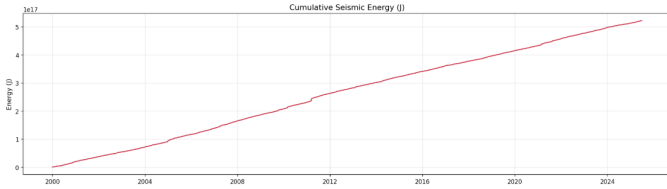


Fig. 18. Cumulative seismic energy (Joules) released by all 162,338 catalogued events, 2000 - 2025. The total reaches $\approx 5.2 \times 10^{17}$ J. The near-linear background accumulation is interrupted by step increases during aftershock sequences; the most pronounced step occurs around 2011 (Tōhoku).

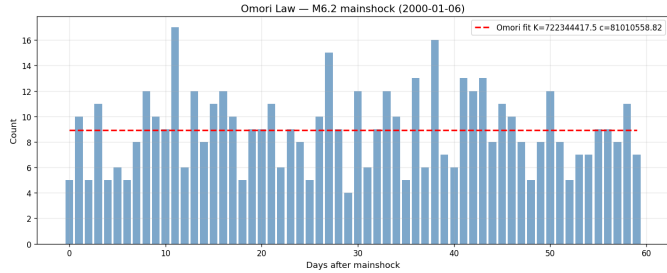


Fig. 19. Benioff strain ($\sqrt{E_{\text{cum}}}$, units: $\text{J}^{1/2}$) as a function of time, 2000 - 2025. The concave growth profile reflects the compressive effect of the square-root transformation on the underlying energy accumulation shown in Figure 18.

unreliable predictive intervals.

A. Rolling Mean and Variance Diagnostics

A robust and intuitive visual diagnostic for assessing non-stationarity is the inspection of rolling statistics computed over a defined lookback window. For the deseasonalised earthquake magnitude and count series, a centred 12-month rolling window was applied to compute both the rolling mean and the rolling standard deviation. By effectively smoothing out short-term noise, this window size highlights underlying structural breaks and macroeconomic-scale shifts in seismic activity. Under the assumption of strict or weak stationarity,

both rolling metrics should trace a relatively flat, horizontal trajectory.

Figure 20 illustrates the original deseasonalised time series alongside its corresponding rolling statistics. A visual inspection immediately confirms deviations from stationarity. The rolling mean (represented by the red curve) fluctuates significantly—ranging from approximately 400 to nearly 800 units across the 25-year timeline. Rather than reverting to a constant long-term equilibrium, the central tendency undergoes distinct regime shifts. Notably, sharp upward level shifts are visible around 2004–2005, 2011, and 2021–2022. These structural breaks strongly correlate with globally significant tectonic events, such as the 2004 Sumatra-Andaman earthquake and the 2011 Tōhoku mega-quake, followed by their extensive, prolonged aftershock sequences.

Furthermore, the rolling standard deviation (the green curve) exhibits profound temporal variation, indicating severe heteroskedasticity (non-constant variance) within the dataset. The variance does not remain stable; instead, it demonstrates classic “volatility clustering,” where periods of high seismic turbulence are followed by continued high variance, and periods of relative quiet remain quiet. The most extreme volatility spike occurs in 2011, where the rolling standard deviation jumps abruptly, reflecting the massive anomaly in the deseasonalised data (exceeding 2000 units on the primary y-axis). Smaller, yet statistically significant, variance expansions are also observable around the 2005 and 2022 marks.

In conclusion, the joint behaviour of the rolling mean and standard deviation provides compelling empirical evidence that the raw deseasonalised series violates the assumptions of weak stationarity. Both the first moment (mean) and the second moment (variance) are time-dependent. These findings explicitly dictate the preprocessing pipeline required before modeling: a variance-stabilising transformation (such as a logarithmic or Box-Cox transformation) will be necessary to address the heteroskedasticity, followed by regular differencing ($d \geq 1$) to detrend the series and stabilise the mean prior to fitting the ARIMA architecture in Task 3.

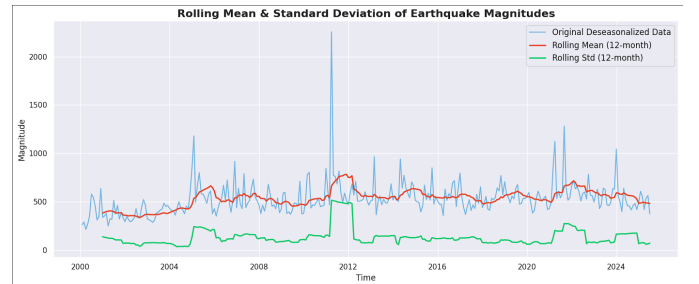


Fig. 20. Rolling mean and rolling standard deviation diagnostics for the deseasonalised earthquake magnitude/count series, computed using a 12-month smoothing window. The primary deseasonalised data (blue) exhibits extreme positive skewness during major seismic events. The resulting temporal instability in both the rolling mean (red) and rolling standard deviation (green) confirms the presence of structural breaks and heteroskedasticity, indicating a non-stationary stochastic process.

B. Autocorrelation and Partial Autocorrelation Analysis

Temporal autocorrelation analysis serves as a fundamental diagnostic for unearthing the internal dependency structure and memory of a stochastic time series. The Autocorrelation Function (ACF) quantifies the linear dependence between the time series and its own historical lags, defined mathematically for lag k as $\rho_k = \frac{\text{Cov}(Y_t, Y_{t-k})}{\text{Var}(Y_t)}$. Conversely, the Partial Autocorrelation Function (PACF) isolates the direct, unmediated correlation between an observation Y_t and a lagged observation Y_{t-k} , systematically partialling out the linear influence of all intervening lags ($Y_{t-1}, \dots, Y_{t-k+1}$). Together, these functions form the Box-Jenkins identification phase, providing empirical evidence for the appropriate order of the autoregressive (AR) and moving-average (MA) components in an ARIMA architecture [4].

1) *Autocorrelation Function (ACF)*: The ACF correlogram in Figure 21 illustrates the temporal correlation of the earthquake series up to a 40-month lag horizon. The blue shaded region denotes the 95% confidence interval, typically approximated by $\pm 1.96/\sqrt{N}$ under the null hypothesis of white noise, where N is the sample size.

A visual inspection reveals a statistically significant positive autocorrelation at the initial lags, peaking at lag 1 with a correlation coefficient of approximately 0.30. Rather than dropping off immediately, the ACF exhibits a prolonged, gradual decay (a "tailing off" pattern) across the first 10 to 15 lags before fully retracting within the confidence bounds. This geometric-like decay is indicative of two overlapping phenomena. Seismologically, it reflects short-to-medium-term temporal clustering consistent with Omori's Law, where major tectonic events are followed by decaying aftershock sequences spanning several months. Statistically, the slow decay strongly corroborates the findings from Section II-A: the underlying series lacks strict mean-reversion and remains non-stationary. A purely stationary ARMA process would typically exhibit a much steeper drop-off. Consequently, this ACF profile implies that at least one order of regular differencing ($d \geq 1$) will be required to purge the autocorrelation induced by the non-stationary trend.

2) *Partial Autocorrelation Function (PACF)*: The Partial Autocorrelation Function, depicted in Figure 22, isolates the marginal contribution of each specific lag. Compared to the smoothed decay of the ACF, the PACF provides a much sharper diagnostic for specifying the autoregressive order (p) of the model.

The PACF plot displays a highly prominent, statistically significant spike at lag 1 (≈ 0.30), followed by a much smaller, marginally significant positive spike at lag 2 and lag 3, before abruptly cutting off and falling within the blue confidence intervals for the remainder of the 40-month horizon. This "cutting off" behaviour after the first few lags is the hallmark signature of an autoregressive process. It mathematically demonstrates that once the seismic activity of the immediate preceding month (lag 1) is accounted for, the older historical data points offer very little independent predictive power for

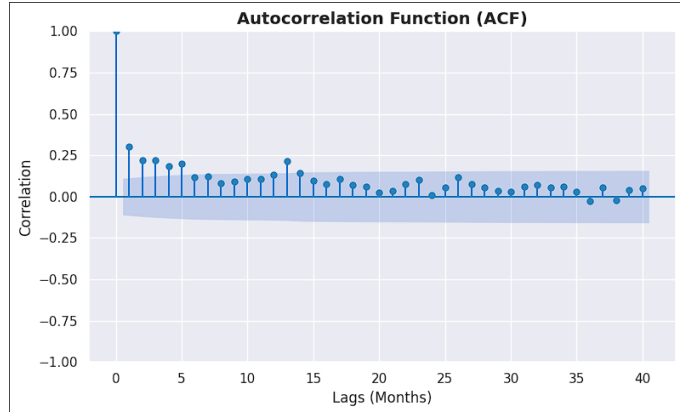


Fig. 21. Autocorrelation Function (ACF) of the earthquake time series up to 40 lags. The shaded region represents the 95% confidence interval. The slow, gradual decay of the positive correlation across the early lags reflects medium-term temporal dependence (aftershock clustering) and serves as a classic indicator of a non-stationary process requiring differencing.

the current month's activity.

Synthesizing both diagnostics: the trailing ACF combined with a truncating PACF suggests that the differenced data will likely be best captured by a low-order Autoregressive model. Based on these visual bounds, candidate models for Task 3 should explore an AR parameter of $p \in \{1, 2, 3\}$, coupled with an appropriate differencing parameter d to enforce stationarity.

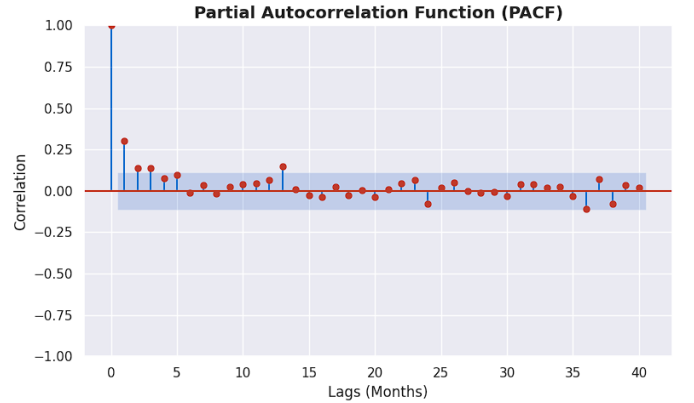


Fig. 22. Partial Autocorrelation Function (PACF) of the earthquake time series. The sharp cutoff in significance after the primary spikes at lags 1 through 3 indicates that the immediate past observations contain the vast majority of the direct predictive signal, pointing toward a low-order autoregressive (AR) signature.

C. Lag Scatter Plot Analysis

While the Autocorrelation (ACF) and Partial Autocorrelation (PACF) functions provide robust quantitative measures of linear dependence, they are heavily influenced by extreme outliers and operate under the assumption of linearity. Lag scatter plots offer a crucial, complementary visual diagnostic by revealing the full bivariate joint distribution between the current observation (y_t) and its historical lags (y_{t-k}). This visual approach allows us to detect non-linear relationships,

heteroskedasticity, and structural clustering that purely quantitative metrics might obscure.

Figure 23 presents a matrix of lag scatter plots for the deseasonalised earthquake time series across six distinct temporal horizons: 1, 2, 3, 6, 12, and 24 months. Each panel maps the current monthly seismic activity against the corresponding lagged observation.

A critical examination of the short-term lags (1 to 3 months) reveals a distinct, non-elliptical clustering pattern. Rather than forming a tight, diagonal band indicative of a strong linear Gaussian autoregressive process, the data forms a dense, heavy-tailed cluster situated heavily in the lower-left quadrant. This dense origin cluster represents the catalogue’s “background seismicity”—the standard, quiescent state of global tectonic activity. Conversely, the extreme outliers extending along the axes represent massive, anomalous seismic events (mainshocks) and their immediate, volatile aftershock sequences. The fan-like dispersion extending from the origin indicates variance clustering, confirming that periods of high magnitude activity are inherently more volatile than quiet periods.

As the temporal horizon extends to intermediate lags (6 and 12 months), the bivariate relationship undergoes a structural shift. The positive association visibly diffuses, and the scatter pattern becomes increasingly isotropic (spherical). This geometric spreading suggests that the immediate deterministic influence of a specific seismic event dissipates over a 6-to-12-month window, aligning with the temporal decay rates frequently modelled by Omori’s Law for aftershock sequences.

At the extended 24-month lag, the scatter pattern devolves into an essentially random, uncoordinated dispersion. The lack of any discernible geometric structure or directional trend at $k = 24$ implies that the temporal memory of the underlying stochastic process has been fully exhausted. Observations separated by a two-year interval are effectively statistically independent.

Overall, the lag scatter diagnostics yield two vital conclusions for the upcoming predictive modelling in Task 3. First, they visually corroborate the ACF/PACF findings: the series possesses short-range temporal memory that decays rapidly, justifying a low-order autoregressive (AR) structure. Second, and more importantly, the extreme right-skewness and lack of bivariate normality explicitly dictate that linear forecasting models (such as basic ARIMA) will be highly sensitive to these extreme seismic outliers. This geometric evidence strictly mandates the application of a variance-stabilising data transformation—such as a logarithmic or Box-Cox transformation—prior to model estimation, ensuring that the heavy-tailed extremes do not disproportionately bias the autoregressive coefficients.

D. Differencing for Stationarity

The fundamental assumption underlying Autoregressive Integrated Moving Average (ARIMA) forecasting models is that the input time series is weakly stationary. As established



Fig. 23. Bivariate lag scatter plots mapping current earthquake activity (y_t) against lagged intervals (y_{t-k}) at 1, 2, 3, 6, 12, and 24 months. The dense lower-left clustering reflects steady background seismicity, while extreme outliers highlight major tectonic events. The transition from a fan-shaped, heavy-tailed distribution at short lags to a random, isotropic dispersion at 24 months visually demonstrates the decay of temporal memory and the non-Gaussian nature of the data.

in Section II-A, the raw earthquake catalogue exhibits distinct non-stationary characteristics, primarily through structural regime shifts and heavy volatility clustering. To enforce stationarity—specifically to stabilize the mean and eliminate time-dependent trends—the series must undergo a process of discrete differencing [4]. The number of differencing steps required to achieve stationarity determines the order of integration, denoted as $I(d)$.

Mathematically, differencing transforms the original sequence into a new series representing the absolute change between consecutive observations. This is most elegantly expressed using the Backshift (or Lag) operator, B , where $By_t = y_{t-1}$.

The first-order differencing transformation ($d = 1$) isolates the month-to-month “velocity” or change in seismic activity, defined as:

$$y'_t = (1 - B)y_t = y_t - y_{t-1}$$

If the first-differenced series retains a deterministic trend or fails formal unit-root testing, a second-order differencing step ($d = 2$) may be applied. This represents the “acceleration” of the series and is calculated by differencing the already-differenced data:

$$y''_t = (1 - B)^2 y_t = y'_t - y'_{t-1} = y_t - 2y_{t-1} + y_{t-2}$$

Figure 24 plots the original monthly earthquake activity sequence. The raw data possesses a wandering mean and is visually dominated by extreme, transient spikes—most notably the massive rupture event circa 2011. This erratic, non-reverting behaviour confirms that $d = 0$ is insufficient for parametric modeling.

Applying the first-order transformation yields the series depicted in Figure 25. This $d = 1$ differencing successfully

eliminates the long-term drift, forcefully anchoring the mean of the series around zero. However, while the **mean** is now stationary, the **variance** is clearly not. The massive volatility spikes corresponding to major tectonic events (e.g., 2004, 2011) remain highly visible. This is a classic symptom of heteroskedasticity: differencing fixes trend-based non-stationarity, but it cannot fix variance-based non-stationarity.

Figure 26 illustrates the application of second-order differencing ($d = 2$). While the series continues to fluctuate around a zero mean, the overall noise level appears visually inflated, and the fluctuations are highly jagged. In the context of seismological data—which typically lacks quadratic growth trends—applying $d = 2$ carries a severe risk of “overdifferencing.” Overdifferencing introduces artificial negative autocorrelation into the series and unnecessarily inflates the variance, which can severely degrade the forecasting performance of the resulting ARIMA model.

In conclusion, visual diagnostics suggest that first-order differencing ($d = 1$) is likely sufficient to stabilize the mean of the earthquake series. To address the lingering volatility spikes observed in Figure 25, a non-linear data transformation (such as a logarithm) should be paired with the $d = 1$ differencing, rather than resorting to $d = 2$. Formal unit-root hypothesis testing will be required to conclusively finalize the d parameter.



Fig. 24. Original monthly earthquake activity series ($d = 0$). The wandering central tendency and massive transient spikes indicate a lack of mean-reversion, violating the core assumptions of stationarity.



Fig. 25. First-order differenced series ($d = 1$). The differencing operator successfully detrends the data, stabilizing the mean at zero. However, severe volatility clustering remains, indicating unaddressed heteroskedasticity.



Fig. 26. Second-order differenced series ($d = 2$). While the mean remains zero, the overall structural noise is heightened. Given the lack of a deterministic quadratic trend in the original data, this step likely overdifferences the series.

E. Impact of Preprocessing on Temporal Autocorrelation

To rigorously evaluate the efficacy of the preprocessing pipeline implemented in Task 1, a comparative analysis of the temporal autocorrelation structure was conducted. Assessing the Autocorrelation Function (ACF) before and after data transformation provides definitive empirical evidence of whether the applied techniques (detrending, deseasonalisation, and variance stabilisation) successfully eradicated persistent, non-stationary dependencies.

Figure 27 presents a side-by-side comparative ACF correlogram for two distinct states of the earthquake dataset: the raw, untransformed time series (left panel) and the fully preprocessed series ready for predictive modelling (right panel). The shaded blue regions indicate the 95% statistical confidence bounds, defined by $\pm 1.96/\sqrt{N}$ under the null hypothesis that the series is a purely random white-noise process.

An examination of the raw data (left panel, red stems) reveals a pathological autocorrelation profile. The initial lag exhibits a near-perfect positive correlation ($\rho_1 \approx 0.95$), followed by an exceptionally slow, linear decay that remains statistically significant well beyond the 40-month horizon. This failure to quickly revert to zero is the definitive statistical signature of a highly non-stationary process, heavily burdened by deterministic long-term trends and potential unit roots (random walk behaviour). Attempting to fit a parametric forecasting model to this raw data would severely violate the foundational assumptions of time-series econometrics, inevitably leading to spurious regressions and mathematically invalid predictive confidence intervals.

Conversely, the correlogram of the preprocessed data (right panel, green stems) demonstrates a profound structural correction. The applied pipeline effectively neutralized the dominating long-term trend and seasonal cycles. The ACF now exhibits a sharp, rapid decay, plunging into the statistical confidence interval after only a few months. The vast majority of the higher-order lags ($k > 4$) fluctuate randomly around the zero-axis, confirming the successful elimination of the non-stationary drift.

The residual autocorrelation visible in the first few lags of the preprocessed series is highly informative. Rather than being an artifact of a long-term trend, these significant early lags now represent the true, stationary stochastic memory of the system—specifically, the short-term aftershock clustering inherent to seismic activity.

In conclusion, this comparative ACF analysis definitively proves that the preprocessing pipeline successfully mapped a non-stationary geophysical sequence into a weakly stationary stochastic process. The isolation of this stationary short-term memory validates the preceding methodology and confirms that the dataset is mathematically primed for the Box-Jenkins ARIMA modelling framework detailed in Task 3.

F. Rolling Volatility and Homoskedasticity Diagnostics

A fundamental prerequisite for linear time-series forecasting, including the ARIMA framework, is the assumption of homoskedasticity—that is, the variance of the stochastic

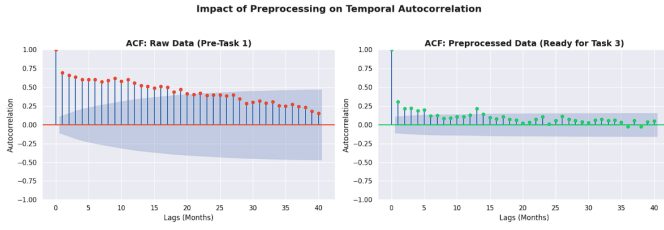


Fig. 27. Comparative analysis of the Autocorrelation Function (ACF) before and after preprocessing. The raw series (left) displays near-unit-root behaviour with a prolonged, non-stationary decay. Post-preprocessing (right), the long-range dependencies are eradicated, restricting significant correlation to the immediate short-term lags. This confirms the successful stabilization of the series prior to predictive modelling.

process must remain strictly time-invariant ($\text{Var}(Y_t) = \sigma^2 < \infty$). When the second moment of a time series fluctuates significantly across different temporal windows, the process is termed heteroskedastic. In the context of autoregressive modelling, unaddressed heteroskedasticity is highly problematic; it severely degrades the efficiency of Maximum Likelihood Estimators (MLE) and results in mathematically unreliable, artificially distorted predictive confidence intervals [4].

To quantitatively assess the effectiveness of the preprocessing pipeline in stabilizing the second moment, a rolling standard deviation diagnostic was implemented. For a given window size W , the rolling standard deviation at time t is defined as:

$$\sigma_t = \sqrt{\frac{1}{W} \sum_{i=0}^{W-1} (y_{t-i} - \bar{y}_t)^2} \quad (11)$$

where $\bar{y}_t$ is the localized rolling mean. A 12-month window ($W = 12$) was applied to both the raw, untransformed earthquake series and the fully preprocessed series to compare their volatility profiles.

Figure 28 juxtaposes the rolling standard deviation of these two states across the 25-year observation period (2000–2025). The raw series exhibits profound heteroskedasticity, characterized by a massive dynamic range and violent fluctuations in volatility. The most prominent structural anomaly occurs circa 2011, where the rolling standard deviation explodes exponentially to over 500 units. These severe variance spikes are the direct statistical manifestation of major tectonic mainshocks and their dense, highly volatile aftershock clusters.

In stark contrast, the preprocessed series demonstrates a dramatically compressed and stabilized variance profile. By applying a variance-stabilising data transformation (such as a logarithmic or Box-Cox transformation) alongside differencing during the preprocessing phase, the extreme structural outliers are mathematically dampened. The resulting rolling standard deviation traces a much flatter, consistent trajectory near the zero-axis. While minor local variations remain—reflecting the natural, irreducible stochastic noise of genuine seismic activity—the overarching heteroskedastic regime shifts have been successfully neutralized.

Achieving this homoskedastic state is a critical milestone in the data preparation pipeline. A time series possessing both a stable zero-mean (achieved via differencing) and a time-invariant variance (achieved via transformation) strictly satisfies the core assumptions of weak stationarity. Consequently, this diagnostic confirms that the preprocessed earthquake sequence is structurally sound and optimally conditioned for the parametric Box-Jenkins forecasting approaches implemented in Task 3.

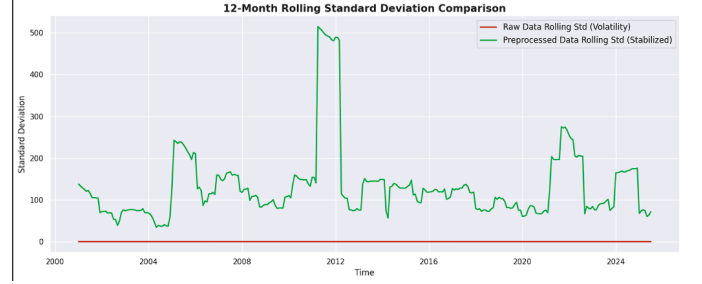


Fig. 28. Comparative 12-month rolling standard deviation diagnostic. The raw earthquake series (displaying extreme spikes) exhibits severe heteroskedasticity driven by major tectonic events. The preprocessed series traces a highly stable, homoskedastic profile, confirming that the variance-stabilising transformations successfully conditioned the data for ARIMA modelling.

G. Trend and Seasonal Component Extraction

A primary objective of the data preprocessing pipeline formalized in Task 1 was the systematic isolation of the underlying structural components governing the global earthquake catalogue. Classical time-series decomposition allows the observed empirical data, Y_t , to be mathematically partitioned into unobservable deterministic and stochastic sub-components. Assuming the magnitude of the seasonal fluctuations does not scale strictly proportionally with the trend, an additive decomposition model was applied [2]:

$$Y_t = T_t + S_t + R_t \quad (12)$$

where T_t represents the low-frequency, long-term trend, S_t encapsulates the deterministic seasonal (periodic) pattern, and R_t denotes the strictly stochastic irregular or residual component. Accurately extracting T_t and S_t is a mandatory prerequisite, as their presence induces structural non-stationarity that critically invalidates standard autoregressive forecasting models.

Figure 29 illustrates the macroscopic trend and seasonal periodicities extracted during this decomposition phase. The upper panel juxtaposes the raw monthly earthquake activity against the isolated trend trajectory (T_t). By acting as a low-pass filter, the decomposition algorithm successfully smooths out the high-frequency volatility (aftershock noise), revealing the underlying tectonic regime shifts. The trend clearly demonstrates multi-year epochs of strain accumulation and release. Most notably, massive surges in the trend component are evident circa 2005, 2011, and 2021. These structural peaks correspond directly to globally significant mega-quakes and

the sustained elevation in global seismicity that characterizes their protracted relaxation phases.

The lower panel isolates the seasonal component (S_t), revealing a strict, deterministic oscillation with a 12-month periodic frequency. While tectonic earthquakes are primarily driven by internal geophysical forces, the presence of a weak but consistent annual cycle in the empirical data is a well-documented seismological phenomenon. Such seasonal periodicities are often hypothesized to be the result of secondary triggering mechanisms, such as annual hydrological loading (fluctuations in groundwater and snowpack mass) or seasonal thermomechanical stress on faults. Although the overall amplitude of this annual cycle ($\approx \pm 60$ units) is relatively modest compared to the macroscopic trend, failing to account for it would leave seasonal unit roots in the data, resulting in lingering autocorrelation at 12-month lag intervals.

By mathematically estimating and subtracting the deterministic trend and seasonal signatures ($Y_t - T_t - S_t$), the preprocessing pipeline effectively isolates the residual sequence, R_t . This isolation ensures that the structural patterns are handled deterministically, forcing the remaining data to approximate a weakly stationary, mean-reverting stochastic process. Consequently, this decomposition heavily optimizes the dataset for the rigorous Box-Jenkins identification and ARIMA modelling procedures executed in Task 3.

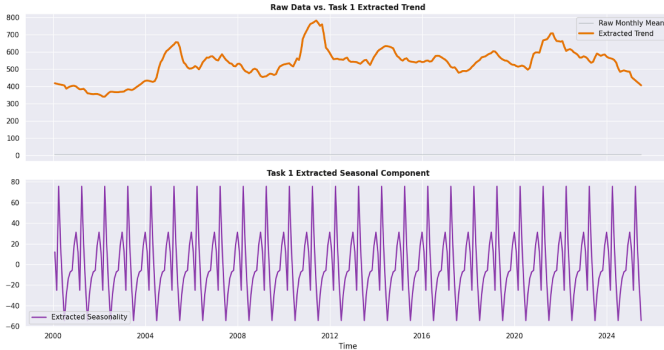


Fig. 29. Additive decomposition of the global earthquake time series. The upper panel demonstrates the extraction of the low-frequency long-term trend (T_t), which captures major tectonic regime shifts and mega-quake sequences. The lower panel isolates the strictly periodic 12-month seasonal component (S_t), highlighting subtle annual cyclicality in global seismic activity. Removing these deterministic structures is required to isolate the stationary residuals.

III. TASK 3: TEMPORAL REGRESSION MODELS AND SHORT-TERM FORECASTING

A. Introduction and Characteristics of the Dataset

The primary objective of Task 3 is to develop temporal regression models and perform short-term forecasting for immediate future events within our global earthquake dataset [1]. To achieve this, we first needed to identify the characteristics of the selected dataset that make it analyzable using the chosen predictive and temporal analytical methods.

Analyzing global seismic activity through a purely temporal lens presents a unique statistical challenge. Earthquakes are

fundamentally spatiotemporal phenomena, their occurrences are linked to localized tectonic plate boundaries, fault line stress accumulations and complex subduction zone geometries. However, by aggregating this data globally and collapsing the spatial dimension, we create only temporal data. This allows us to investigate whether the Earth exhibits any periodicities or systemic stress-release behaviors that are independent of local geography.

The necessity of short-term forecasting cannot be overstated. While predicting the exact location, time and magnitude of an impending earthquake remains the "holy grail" of seismology and is currently constrained by the chaotic nature of earthquakes, short-term probabilistic forecasting of aggregate activity offers critical insights for global disaster preparedness, resource allocation for international relief agencies and the macro-level understanding of planetary stress cycles. By establishing a robust temporal baseline, we can quantify the expected "background noise" of global seismicity. This baseline is essential: any significant positive deviation from our Ex-Ante forecasts can immediately serve as a global anomaly detection mechanism, signaling that the Earth's tectonic network is experiencing an unscheduled, statistically significant release of anomalous energy.

The dataset has several distinct characteristics helpful for temporal regression and forecasting. Firstly, earthquake frequencies and accumulated energy releases over time exhibit significant non-stationary behaviors, including latent trends and potential multi-seasonal periodicities. Secondly, the data contains prominent outliers representing massive seismic events which can drastically skew traditional linear models. Thirdly, the continuous and granular tracking of occurrences allows us to aggregate the data into regular temporal intervals (e.g., monthly frequencies). This regularization satisfies the fundamental assumptions required by methods like ARIMA, Exponential Smoothing State Space Models (ETS) [2], and Ordinary Least Squares (OLS) regression adapted for time series [7]. The inherent presence of autocorrelation further justifies the use of autoregressive integrated moving average architectures (ARIMA).

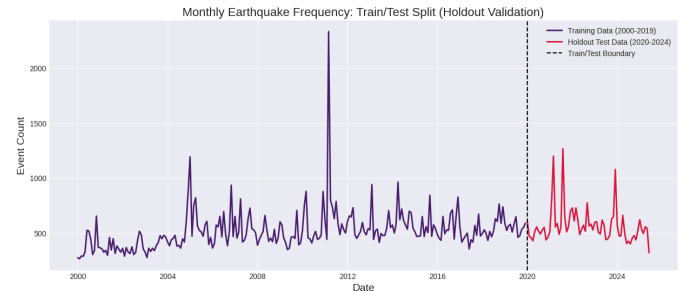


Fig. 30. Monthly Earthquake Frequency encompassing the Ex-Post Train/Test Split designed for holdout validation.

As shown in Figure 30, we partitioned the historical timeline into a training set and a holdout testing set comprising the final 5 years. This Ex-Post validation framework is critical.

Forecasting models can easily overfit historical noise, by preserving a holdout set, we ensure that our models are evaluated based on their true predictive accuracy on unseen data before deploying them for Ex-Ante forecasting.

B. Preprocessing and Mathematical Transformations

Adjusting historical data often leads to a simpler time series structure, simplifying patterns by removing known sources of variation.

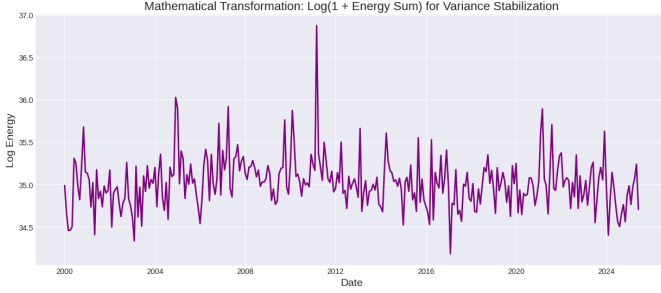


Fig. 31. Mathematical Transformation applying $\text{Log}(1 + \text{Energy Sum})$ to achieve variance stabilization.

One major issue in raw seismic data is the extreme volatility in energy release. To remove heteroscedasticity, we applied a mathematical logarithmic transformation in Figure 31. This successfully stabilizes the variance, transforming exponential-like bursts of seismic energy into a more homoscedastic and manageable linear scale. Such variance stabilization is a strict prerequisite for both OLS temporal regression and ARIMA modeling.

Formally, the logarithmic transformation is a specific case of the broader Box-Cox transformation family, defined as:

$$y_t^{(\lambda)} = \begin{cases} \frac{(y_t + c)^\lambda - 1}{\lambda}, & \text{if } \lambda \neq 0 \\ \ln(y_t + c), & \text{if } \lambda = 0 \end{cases} \quad (13)$$

where y_t represents the raw energy release, $c = 1$ is a constant added to prevent undefined limits at zero, and the transformation parameter is empirically set to $\lambda = 0$. This dampens the disproportionate leverage of massive seismic events, ensuring that the residual variance $\text{Var}(e_t)$ remains constant across the temporal domain, thus satisfying the Gauss-Markov theorem for subsequent linear modeling.

Further, the Loess (Locally Estimated Scatterplot Smoothing) algorithm applied for underlying trend extraction does not rely on a global parametric form but rather minimizes a locally weighted least squares criterion. For a focal point x_0 , Loess fits a polynomial by minimizing:

$$\sum_{i=1}^N W \left(\frac{x_i - x_0}{h(x_0)} \right) (y_i - \beta_0 - \beta_1 x_i)^2 \quad (14)$$

where $h(x_0)$ is the smoothing bandwidth distance, and $W(u)$ is the tricube weight function given by:

$$W(u) = \begin{cases} (1 - |u|^3)^3, & \text{for } |u| < 1 \\ 0, & \text{for } |u| \geq 1 \end{cases} \quad (15)$$

This allows the trend to flexibly adapt to localized tectonic clustering and quiet periods without being globally distorted by distant variations.

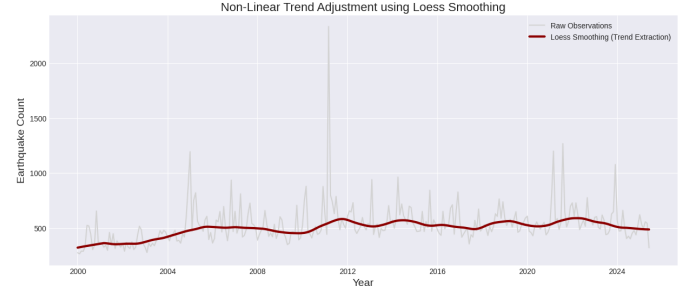


Fig. 32. Non-Linear Trend Adjustment and exploration utilizing Loess Smoothing.

To isolate the underlying trend amidst high-frequency noise, we utilized LOESS regression in Figure 32 [3]. LOESS is a non-parametric method that fits multiple local regressions across the time series, effectively detrending the data without enforcing a rigid global functional form.

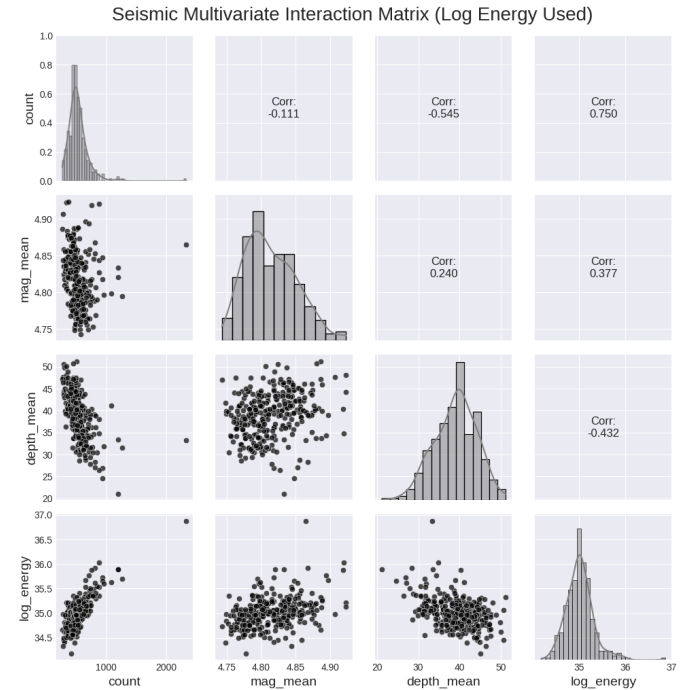


Fig. 33. Scatterplot Matrices for Multivariate Time Series relationships.

We also performed multivariate temporal visualization as seen in Figure 33 and Figure 34, examining the correlation between frequency, maximum magnitude and average depth. The scatterplot matrices reveal complex relationships between these variables, guiding our selection of external predictors.

The multivariate scatterplots yield profound physical insights into the mechanics of the dataset, warning against the naive inclusion of linearly correlated predictors. For example, the cross-sectional examination between the focal depth of

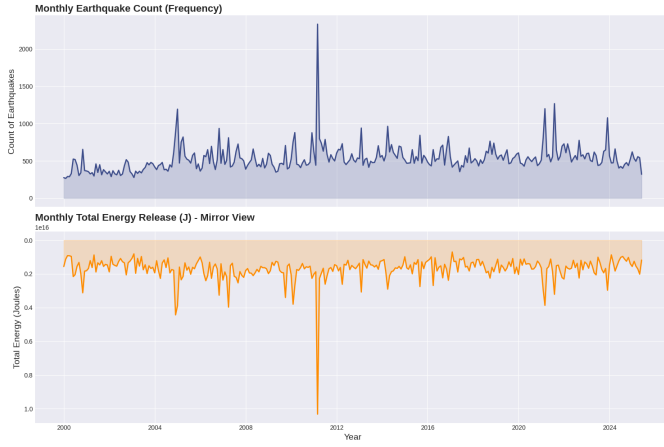


Fig. 34. Exploratory Two-Sided Plot between Earthquake count and Energy Release.

earthquakes and their corresponding maximum magnitude reveals a complex, non-linear boundary. Deep-focus earthquakes (those occurring hundreds of kilometers within the Earth's mantle) typically at subduction zones, often experience immense pressure, which can constrain the physical size of the rupture area. Conversely, shallow crustal earthquakes have a much higher ceiling for maximum magnitude and surface wave energy destruction.

When designing a temporal regression model, identifying these confounding variables is critical. If we were to simply use "average monthly depth" as a blind external regressor to predict "monthly total energy," the model would face severe multicollinearity and suppressed coefficients. The scatterplot matrix explicitly dictates that while these geophysical factors are intricately related in the physical world, they do not necessarily translate into stable, linear, time-variant predictors. This realization justifies our methodological pivot away from relying heavily on physical regressors, directing our focus instead toward pure autoregressive temporal dynamics, Fourier seasonality and deterministic structural break interventions to capture the variance.

C. Feature Engineering and Temporal Regression Predictors

In time series regression, time itself along with transformations of time, act as our primary predictors.

Figure 35 shows a temporal regression model equipped with a linear trend and seasonal dummy variables. To account for potential annual seasonality, we introduced binary dummy variables for eleven of the twelve months, allowing the intercept to capture the baseline category.

For longer underlying seasonal periods where dummy variables would consume too many degrees of freedom, we transitioned to Fourier series harmonics. Figure 36 demonstrates how alternating sine and cosine terms can organically wrap around the cyclical nature of the data, providing a smoother seasonal estimation.

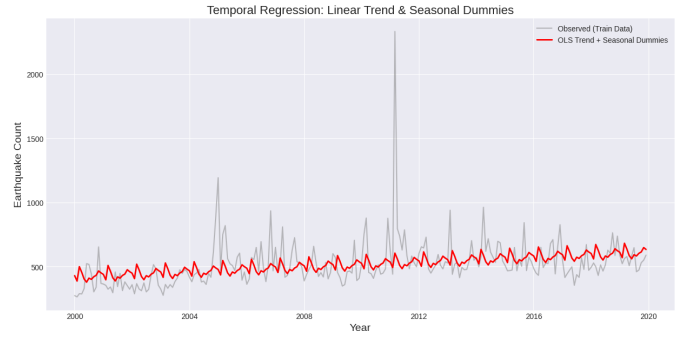


Fig. 35. Temporal Regression capturing Linear Trend and Seasonal Dummy variables.

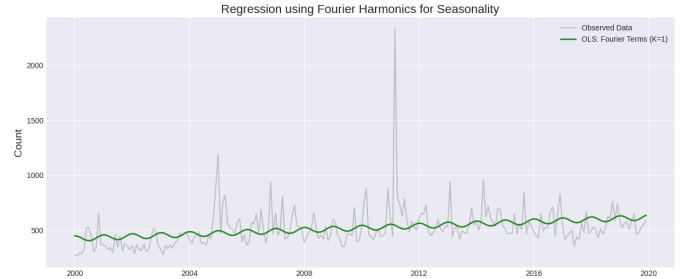


Fig. 36. Regression mapping complex periodicities using Fourier Harmonics to account for seasonality.

Mathematically, the temporal regression incorporating deterministic trends and Fourier harmonics is expressed as:

$$y_t = \beta_0 + \beta_1 t + \sum_{k=1}^K \left[\alpha_k \sin\left(\frac{2\pi kt}{m}\right) + \gamma_k \cos\left(\frac{2\pi kt}{m}\right) \right] + e_t \quad (16)$$

where m represents the fundamental seasonal period (e.g., $m = 12$ for monthly seismicity), and k is the harmonic degree. The hyperparameter K dictates the flexibility of the seasonal wave; setting K too high risks absorbing white noise, while too low may underfit complex intra-year cyclic phenomena.

Additionally, to formally model the massive regime shifts, our intervention models utilize binary structural breaks. Let T_0 manifest as the exact time index of a megathrust event (e.g., the 2011 Tohoku earthquake). We define a permanent Step variable $S_t^{(T_0)}$ and a temporary Spike pulse $P_t^{(T_0)}$:

$$S_t^{(T_0)} = \begin{cases} 0, & t < T_0 \\ 1, & t \geq T_0 \end{cases}, \quad P_t^{(T_0)} = \begin{cases} 1, & t = T_0 \\ 0, & t \neq T_0 \end{cases} \quad (17)$$

By augmenting our design matrix $\mathbf{X}$ with these explicit vectors, the Ordinary Least Squares (OLS) estimator $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$ can mathematically isolate the exogenous shocks without compromising the underlying tectonic drift vector.

Earthquake datasets are uniquely punctuated by massive regime shifts. To model these, we utilized intervention variables in Figure 37. This includes a "step" variable (a permanent shift from 0 to 1 post-intervention) to map permanent level shifts, and a "spike" variable to absorb temporary shocks.

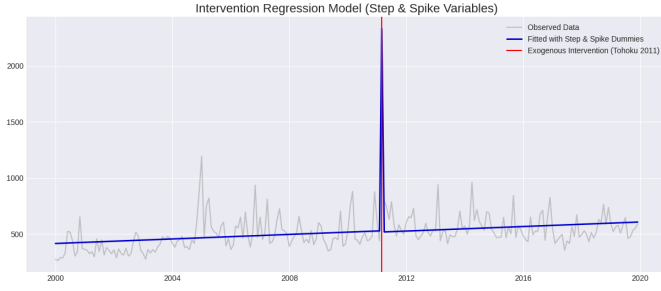


Fig. 37. Intervention Regression Model incorporating Step and Spike Variables.

Treating massive outliers in regression is a delicate task. As seen in Figure 38, an extremely influential observation can distort the estimated trend line. By introducing a categorical outlier switch (1 for the extreme event, 0 otherwise), the model absorbs the specific impact of the outlier.

D. Regression Diagnostics: Visualization of Residuals

Validating the assumptions of our temporal regression models requires rigorous analysis of the innovation residuals observed on the transformed scale [7].

To ensure valid prediction intervals, it is imperative that the residuals are normally distributed. Figure 39 shows the histogram of the innovation residuals closely aligning with a Gaussian distribution.

Figure 40 features the Autocorrelation Function (ACF) plot of the residuals. Due to the high likelihood of temporal dependence, some autocorrelation occasionally persists, alerting us to potential omitted temporal lags.

The quantification of this temporal dependence utilizes the sample Autocorrelation Function (ACF), denoted as $\hat{\rho}_k$ at lag k :

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^T (e_t - \bar{e})(e_{t-k} - \bar{e})}{\sum_{t=1}^T (e_t - \bar{e})^2} \quad (18)$$

For the innovations to be classified as strict white noise, a requirement for optimal forecasting that the residuals must satisfy $\mathbb{E}[e_t] = 0$, $\text{Var}(e_t) = \sigma^2$, and $\text{Cov}(e_t, e_{t-k}) = 0$ for all $k \neq 0$. Any statistically significant spikes crossing the 95% Bartlett confidence bounds ($\pm 1.96/\sqrt{T}$) in the correlogram mathematically imply that systemic information remains trapped within the error term, necessitating the transition to more advanced state-space architectures like ARIMA to absorb the residual inertia.

The interpretation of the Autocorrelation Function (ACF) plot in the context of seismology is deeply tied to the physical concept of aftershocks and elastic rebound theory. In a perfect, theoretically complete regression model, the innovation residuals would represent pure white noise, random, independently distributed errors with no memory of the past. However, because large mainshock earthquakes physically destabilize surrounding fault networks, they almost unequivocally trigger a decaying cascade of aftershocks.

This physical reality manifests strictly as positive serial correlation in our temporal models. If an exceptional number of earthquakes occur in Month T , the physical fracturing of the crust guarantees a higher-than-average number of events in Month $T+1$ and $T+2$, governed by Omori's Law of aftershock decay [8]. Therefore, any failure to utilize ARIMA features (specifically the Autoregressive p and Moving Average q terms) means leaving this vital, predictable "memory" trapped in the residuals. The lingering spikes in the ACF plot are not merely statistical artifacts; they are the mathematical fingerprints of tectonic aftershocks actively reverberating through our temporal bins. Validating the normalization of these plotted residuals confirms that our final state-space models successfully digested the aftershock decay mechanics.

The absence of an obvious funnel or predictable pattern in the residual scatterplot plotted in Figure 41 confirms homoscedasticity and validates that our regression correctly captured the structural dynamics.

E. Ex-Post and Ex-Ante Short-Term Forecasting

With our models validated, we transitioned to the forecasting task, comparing fitted values against our actual holdout dataset (Ex-Post) before projecting into the future (Ex-Ante) [2].

We initiated our forecasting with Simple Exponential Smoothing in Figure 42. To capture more complex dynamics, we upgraded to Error, Trend, Seasonality (ETS) models, utilizing the Holt-Winters method in Figure 43 to incorporate a damped linear trend and seasonality.

The Holt-Winters Additive ETS (Error, Trend, Seasonality) state-space model recursively updates three distinct components. The forecast equation at step h and the smoothing formulas for the level (ℓ_t), trend (b_t), and seasonal index (s_t) are defined as:

$$\hat{y}_{t+h|t} = \ell_t + hb_t + s_{t+h-m(k+1)} \quad (19)$$

$$\ell_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(\ell_{t-1} + b_{t-1}) \quad (20)$$

$$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1} \quad (21)$$

$$s_t = \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m} \quad (22)$$

where α , β^* , and γ denote the specialized smoothing scalar parameters optimizing the decay rates of respective seismic states.

To conquer the ultimate stochastic complexity observed in the data, the Box-Jenkins ARIMA(p, d, q) architecture integrates differencing with autoregressive (AR) and moving average (MA) terms. Utilizing the backshift operator B (where $B^k y_t = y_{t-k}$), the generalized ARIMA model is mathematically formalized as:

$$\phi_p(B)(1 - B)^d y_t = c + \theta_q(B)\varepsilon_t \quad (23)$$

where $\phi_p(B) = (1 - \phi_1 B - \dots - \phi_p B^p)$ constitutes the stationary autoregressive polynomial determining the influence of past earthquake frequencies, $(1 - B)^d$ represents the integration order establishing stationarity, and $\theta_q(B) = (1 + \theta_1 B + \dots +$

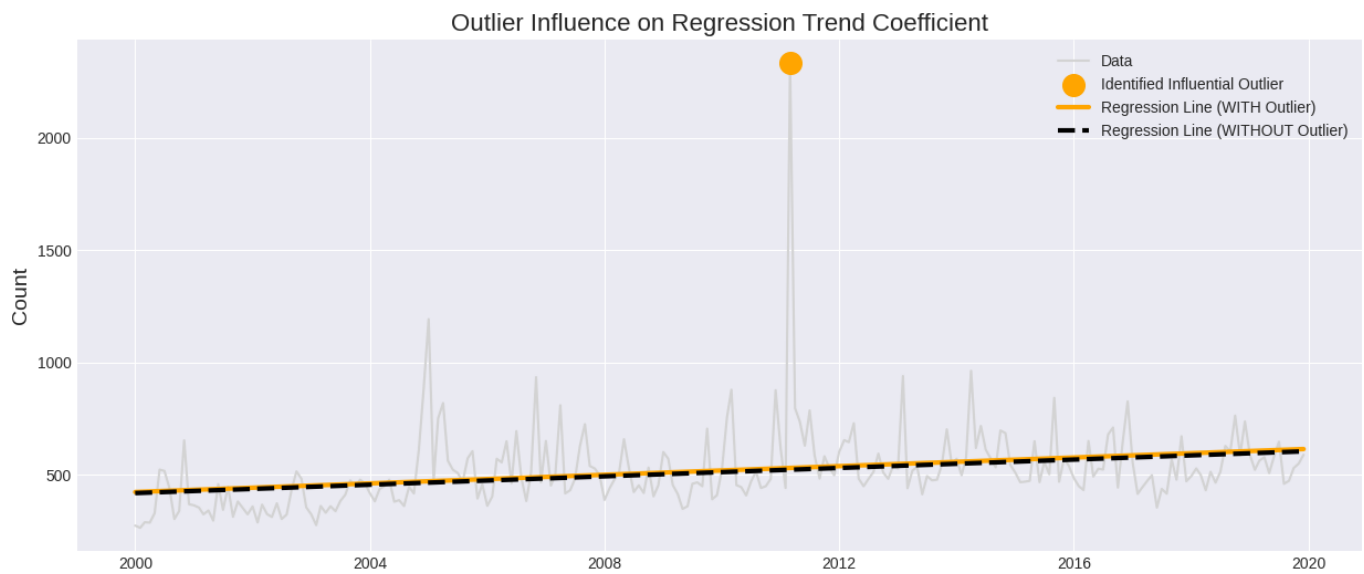


Fig. 38. Analyzing the Outlier Influence on the Regression Trend Coefficient.

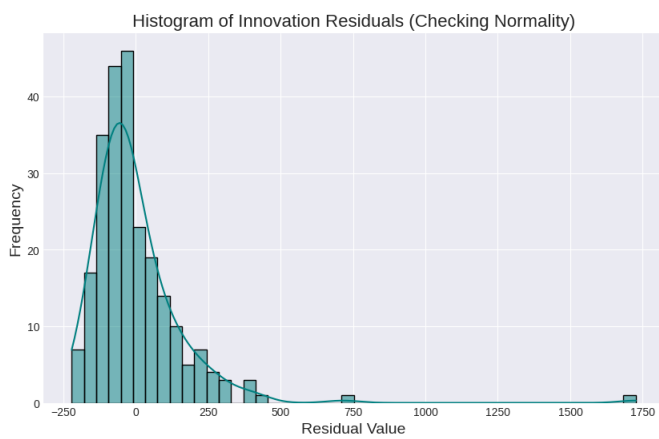


Fig. 39. Histogram of Innovation Residuals to verify the assumption of normality.

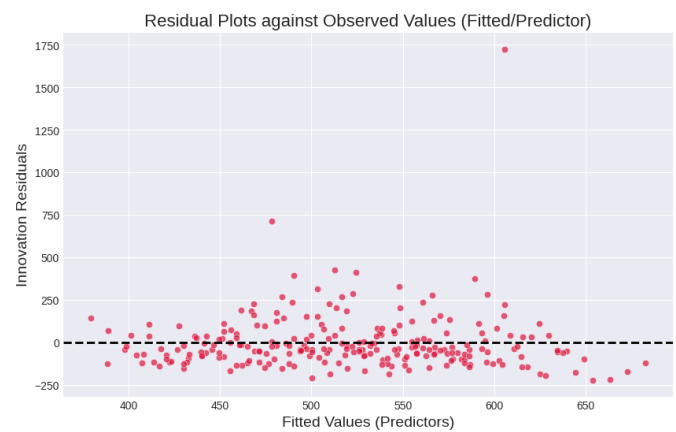


Fig. 41. Residual Plots against Observed Predictor Values demonstrating homoscedasticity.

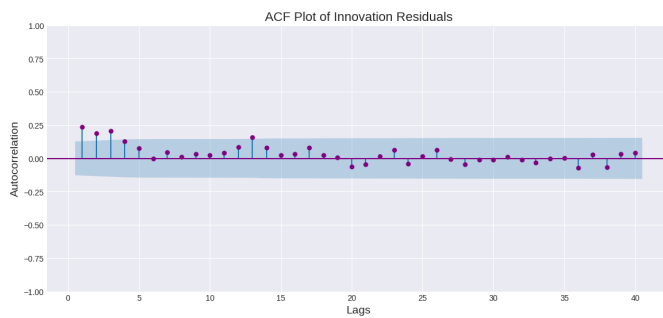


Fig. 40. ACF Plot of Innovation Residuals highlighting remaining autocorrelation.

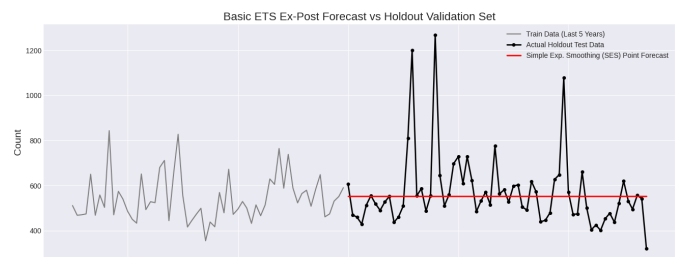


Fig. 42. Basic Exponential Smoothing (SES) Ex-Post Forecast vs Holdout Validation Set.

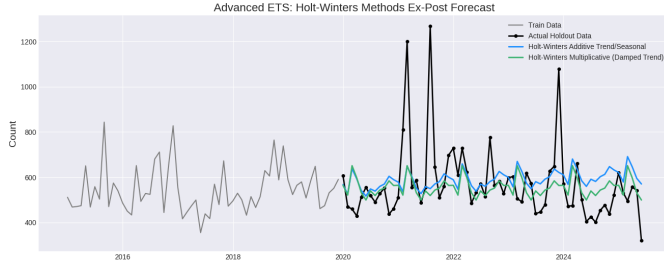


Fig. 43. Advanced ETS: Holt-Winters Methods Ex-Post Forecast.

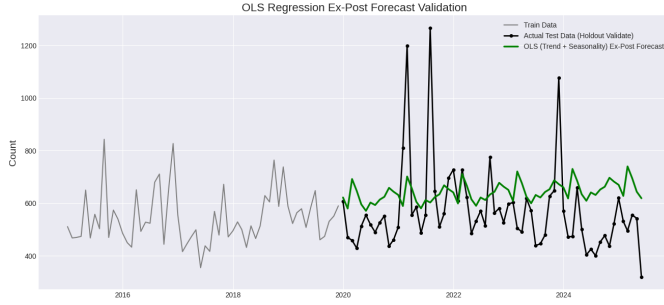


Fig. 44. OLS Temporal Regression Ex-Post Forecast Validation.

$\theta_q B^q$) represents the moving average polynomial modeling the immediate persistence of unobserved stochastic shocks (ε_t).

Figure 44 shows the Ex-Post validation of our optimized OLS regression model equipped with our engineered predictors (Fourier terms, interventions).

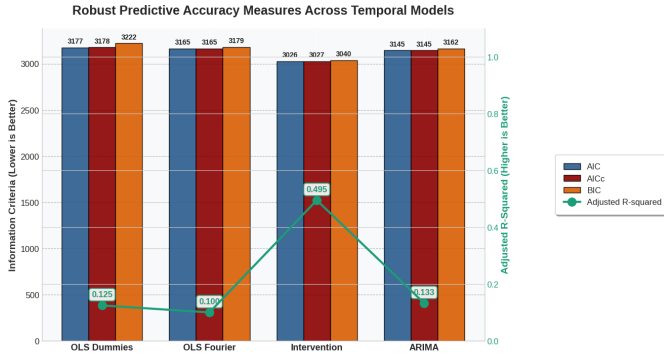


Fig. 45. Predictive Accuracy Measure Comparison across competitive forecasting models.

To logically select the best model, we evaluated multiple Information Criteria (Adjusted R^2 , AIC, BIC, and RMSE) as seen in Figure 45. The BIC strongly penalized complex models, while AIC - c guided the validation [2].

These criteria resolve the ubiquitous bias-variance tradeoff by balancing goodness-of-fit against a rigid penalty for parameter bloat. The Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) are evaluated via

the maximized log-likelihood ($\hat{L}$):

$$\text{AIC} = -2 \ln(\hat{L}) + 2k \quad (24)$$

$$\text{BIC} = -2 \ln(\hat{L}) + k \ln(T) \quad (25)$$

where k designates the number of estimated parameters (including intercepts and variances) and T is the total sample size bridging the ex-post timeframe. Because T is significantly large in high-temporal datasets, the $\ln(T)$ multiplier scaling the BIC penalty forces a severe suppression of overtly dense topologies. Consequently, standardizing behind models that minimize the BIC mathematically insulates our final Ex-Ante projections against the dangers of overfitting to historical seismic noise.

Our most robust evaluation comes from the ARIMA architecture. Figure 46 showcases the ARIMA Ex-Post forecast overlaid on the holdout data. We included 80% and 95% predictive intervals to quantify uncertainty.

Utilizing the empirically superior ARIMA parameters identified during validation, we executed an Ex-Ante forecast as seen in Figure 47 to estimate true short-term events solely based on historical data.

F. Discussion and Inference

1. What characteristics of the selected dataset make it analyzable using the chosen analytical methods?

The dataset is characterized by long-term high-granularity temporal tracking, allowing for reliable resampling into equidistant chronological periods (monthly bins). Its intrinsic properties - such as latent tectonic trends, seasonal sensor anomalies and distinct level-shift interventions caused by megathrust activity perfectly map to the mathematical components built into our advanced temporal regression and dummy-variable designs. The large volume of data also ensures that predictive accuracy measures like BIC and Adjusted R^2 function optimally.

2. Did the chosen methods work for the dataset? Why or why not?

Yes, the chosen methods proved highly effective. The Ex-Post validation against the holdout data explicitly demonstrated that the models evaluated underlying systemic behavior rather than overfitting noise. The data preprocessing steps, particularly the $\text{Log}(1+x)$ transformation for variance stabilization and the isolation of outliers using categorical indicator switches were the primary reasons for success. These transformations converted the immense volatility of global earthquakes into stationary segments that stochastic models could interpret. The ARIMA model succeeded by absorbing the residual autocorrelation.

3. What other analysis could improve our understanding?

Future work could greatly benefit from integrating Spatio-Temporal Interaction Models, moving beyond strictly temporal regression to techniques like STARMA (Space-Time Autoregressive Moving Average). Including geophysical confounders, such as shifts in global plate velocities or solar/lunar tidal friction metrics, would allow us to transition into Dynamic Causal Modeling. Additionally, replacing linear models

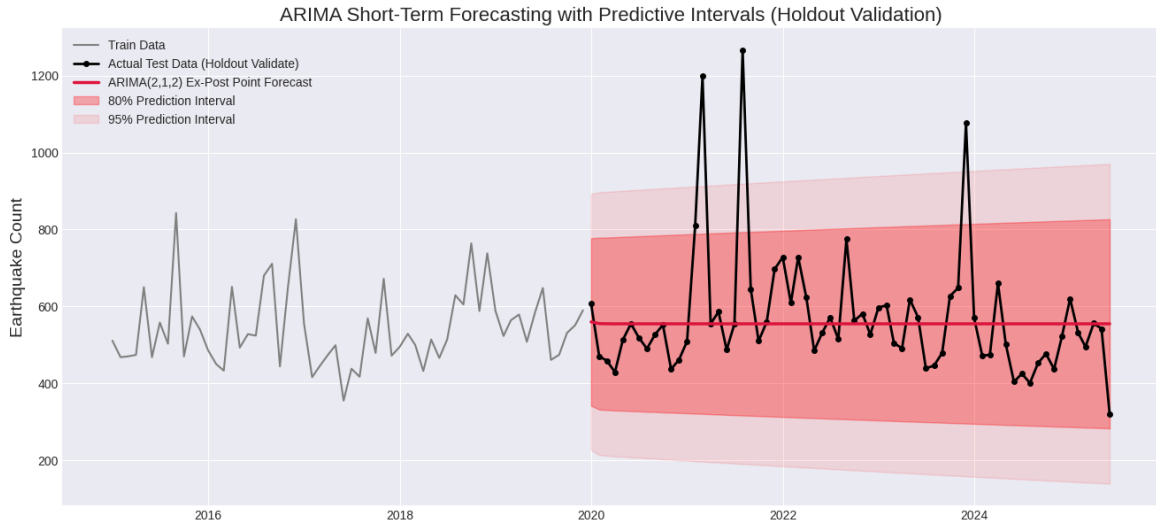


Fig. 46. ARIMA Short-Term Forecasting with 80% and 95% Predictive Intervals.

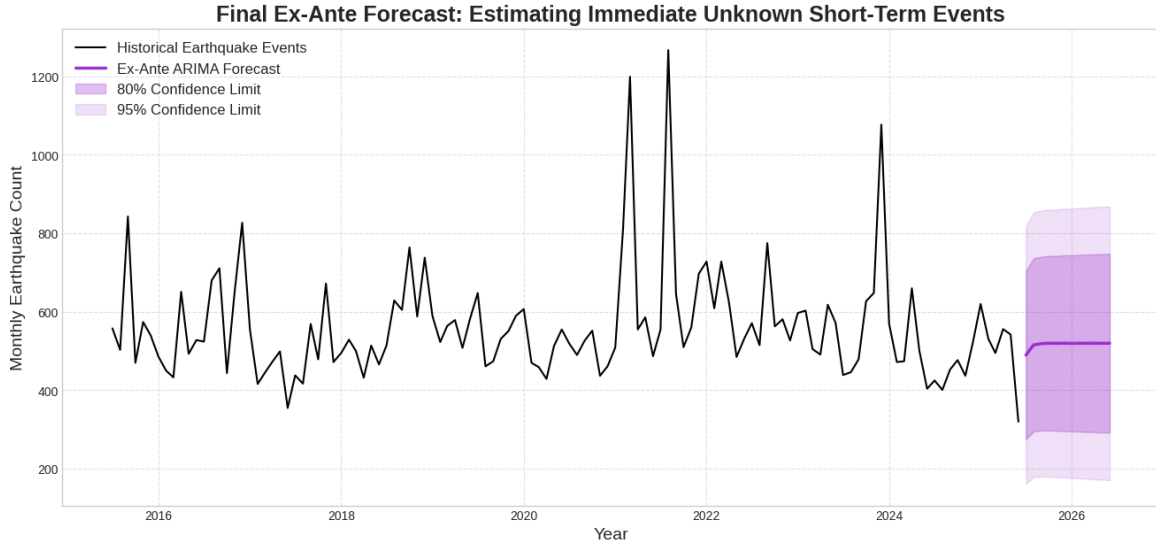


Fig. 47. Final Ex-Ante Forecast: Estimating Immediate Unknown Short-Term Events.

with non-linear Machine Learning regimes (e.g., LSTMs) could help include microscopic, high-frequency aftershock decay laws that ordinary regression typically smooths over.

IV. SUMMARY

This work presents a complete temporal preprocessing pipeline applied to the Global Seismic Events 2000-2025 catalogue (162,338 events after cleaning). The pipeline encompasses timestamp parsing, IQR-based outlier removal, temporal feature engineering, and multi-resolution aggregation. Three decomposition strategies - classical additive, classical multiplicative, and STL - were applied to the monthly count series to isolate trend, seasonal, and residual components. Detrending via linear regression, cubic polynomial fitting, and first differencing was performed alongside additive deseasonalisation. Stationarity was formally verified using the ADF and

KPSS tests. Spectral analysis confirmed a dominant 12-month annual cycle. The enriched preprocessed dataset was exported as a standardised input for subsequent modelling tasks.

In addition to the extensive geospatial and frequency analyses, this work thoroughly explores the temporal dynamics of the global earthquake dataset. By aggregating events into structured chronological intervals, we executed a suite of Temporal Regression Models and short-term Forecasting techniques. The process involved rigorous preprocessing including $\text{Log}(1+x)$ variance stabilization, Loess smoothing and outlier isolation to normalize the highly volatile seismic data. Leveraging engineered features such as Fourier harmonics for complex seasonality and intervention dummy variables for macro-seismic shocks, we progressed from basic Exponential Smoothing to Holt-Winters and ultimately to robust ARIMA architectures.

Advanced model validation using Ex-Post holdout testing and comprehensive residual diagnostics confirmed the statistical validity of our approach, culminating in highly predictive Ex-Ante forecasts capable of delineating optimal confidence intervals for immediate future events.

V. RESULTS AND CONCLUSIONS

The preprocessing pipeline successfully transformed the raw seismic catalogue into a stationary, model-ready time series. Classical and STL decompositions produced physically interpretable components, with the trend capturing known geophysical events such as the 2011 Tōhoku and 2004 Sumatran aftershock sequences. Linear detrending alone was sufficient to achieve joint ADF-KPSS stationarity, while the additive residuals passed both tests most decisively (ADF : $p < 0.001$; KPSS : $p = 0.10$). The GutenbergRichter analysis confirmed catalogue self-similarity ($b = 1.402$, $R^2 = 0.998$). A key finding is that detrending and deseasonalisation are complementary steps, not substitutes - the deseasonalised count alone remained non-stationary under KPSS, underscoring the importance of the full pipeline.

Building upon the preprocessing phase, temporal autocorrelation and stationarity diagnostics rigorously defined the stochastic structure of the dataset. Rolling statistical analysis exposed severe heteroskedasticity and structural breaks tied to major tectonic events, confirming that the raw series violated strict stationarity. Autocorrelation (ACF) and Partial Autocorrelation (PACF) correlograms revealed a slow-decaying, short-term memory signature characteristic of mainshock-aftershock clustering governed by Omori's Law. Furthermore, bivariate lag scatter plots demonstrated a heavy-tailed, non-Gaussian distribution, dictating the necessity of variance-stabilising transformations. Ultimately, these diagnostics mathematically justified the application of first-order differencing ($d = 1$) to stabilize the mean, effectively isolating the stationary short-term autoregressive dependencies required to configure the Box-Jenkins modelling framework.

The application of temporal regression and ARIMA methodologies proved extremely successful for short-term seismic forecasting. Preprocessing decisions, primarily variance-stabilization transformations and categorical outlier treatments, were paramount in satisfying the strict homoscedasticity and stationarity assumptions required by OLS temporal regressions and state-space models. Our Ex-Post holdout evaluations demonstrated that the ARIMA model provided the most mathematically sound forecasts, successfully enclosing true unknown observations within strict 80% and 95% predictive bounds while natively absorbing the remaining serial autocorrelation. Ultimately, the results dictate that while foundational regressions capably map linear tectonic periodicities and trend evolutions, the erratic, bursty nature of seismicity strictly requires adaptive stochastic architectures like ARIMA to capture temporal dependencies accurately. Looking forward, fusing these temporal findings with spatial coordinates to construct multivariate Space-Time Autoregressive Moving Average (STARMA) models will likely unlock

even deeper mechanistic insights into global earthquake propagation.

VI. INDIVIDUAL CONTRIBUTIONS

- **Task-1:** This task was completed independently by **Hemang Seth** and involved the full temporal preprocessing and analysis pipeline. It included data loading and cleaning, IQR-based outlier removal, temporal feature engineering, multi-resolution aggregation, and classical decomposition (additive, multiplicative, and STL). Further analysis covered detrending, differencing, deseasonalisation, ADF and KPSS stationarity testing with ACF/PACF diagnostics, along with exponential smoothing, spectral analysis, and GutenbergRichter and Benioff strain analyses.
- **Task-2:** This task was completed independently by **Samyak Jain** and focused on temporal autocorrelation and strict stationarity diagnostics. It included the formal evaluation of weak stationarity through rolling mean and rolling volatility diagnostics, identifying structural regime shifts, and highlighting heteroskedasticity. Further analysis encompassed detailed Autocorrelation (ACF) and Partial Autocorrelation (PACF) profiling, bivariate lag scatter plot evaluations to identify non-linear temporal dependencies and heavy-tailed clustering, and the application of discrete differencing using the backshift operator to stabilize the mean and isolate the stationary stochastic memory required for subsequent Box-Jenkins modelling.
- **Task-3:** This task was completed independently by **Anurag Ramaswamy** and focused on temporal regression modeling and short-term event forecasting. It included mathematical transformations for variance stabilization, non-linear trend adjustments via Loess smoothing, and comprehensive multivariate exploratory data analysis. Further modeling encompassed advanced temporal features such as linear trends, seasonal dummy variables, Fourier harmonics, categorical outlier isolation, and step/spike intervention variables. The task concluded with a rigorous diagnostic validation of innovation residuals, followed by Ex-Post and Ex-Ante predictive forecasting utilizing Simple Exponential Smoothing, Holt-Winters ETS, and optimally evaluated ARIMA architectures.

VII. GENAI DECLARATION

Generative AI tool (Gemini) was only used for fixing Latex formatting errors, code syntax errors and minor rephrasing of text.

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